

Teaching Cø / N*: A Graduate Instructor's Manual

*Lecture notes, comparative theory placement, validation arguments, and
course materials for the network-dynamics theory of phenomenal
consciousness*

Based on the Cø / N* research program of Phil Stilwell

Instructor edition - July 18, 2026

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Preface

This manual is designed to help an instructor teach a new theory without either oversimplifying it or treating its central claims as already established. The CØ / N* research program proposes a precise candidate account of minimal phenomenal presence and then surrounds that core claim with methodological disciplines: fix the candidate system before measuring it; distinguish causal availability from report; separate scientific indeterminacy from invalid inference; audit every conjunct against simpler rivals; extend presence to phenomenal character only through an independently tested bridge; and explain what it would mean for the model to become structurally intelligible.

The intended audience is graduate students in philosophy of mind, consciousness science, cognitive neuroscience, complex systems, artificial intelligence, or philosophy of science. The register assumes intellectual maturity but not prior allegiance to a particular consciousness theory. Mathematical expressions are used to make commitments visible. Every formal claim should be translated into plain language, linked to a scientific purpose, and paired with a failure condition.

The manual supports a 14-session seminar. A shorter course can use Sessions 1, 2, 3, 5, 7, 8, 9, 10, and 13 as a nine-session core. A four-session workshop can use Sessions 1-2, 3-6 as a single component session, 7-8, and 9-14 as a synthesis session. Each full session contains:

- learning objectives;
- instructor preparation;
- a timed teaching plan;
- lecture notes written for oral delivery;
- a board or display plan;
- a slide sequence;
- an active-learning exercise;
- discussion prompts;
- common misconceptions and corrective responses;
- a short assessment or exit ticket; and
- assigned and optional readings.

The manual treats the seven papers as one research constellation rather than seven competing statements. The core paper supplies the central hypothesis. The companion papers specify the bearer, one difficult component, the evidence policy, the component audit, the character extension, and the model's explanatory form.

Instructor stance

Teach the theory in the conditional and adversarial mood. Students should be able to state what the model claims, why each element is included, what would count for and against it, and where its present evidence is incomplete. A successful course does not produce automatic agreement with the theory. It produces readers capable of evaluating it fairly and designing studies that could genuinely change its status.

Use four recurring questions:

1. What is the explanandum?
2. What is the candidate bearer?
3. What evidence would license the inference?
4. What observation would force revision?

Notation policy

The papers use both C_0 and C_0 for minimal phenomenal presence. This manual uses C_0 in prose and C_0 in formulas. The three components are written N_1 , N_2 , and N_3 . The bundle is $N^* = N_1 \wedge N_2 \wedge N_3$. Viability is V . A positive classification is never treated as a direct observation of another subject's phenomenology; it is a licensed scientific attribution under a declared model and validity envelope.

Part I. Course Architecture

1. The theory in one page

The target is **minimal phenomenal presence**: the fact that there is something it is like for a candidate system to undergo a state. The target does not require verbal report, explicit attention, intelligence, self-reflection, autobiographical memory, or metacognitive ownership.

The core conjecture is:

$$H_{C_0}: V(S; R, \Delta) \wedge (N_1(S, c) \wedge N_2(S, c) \wedge N_3(S, c)) \Leftrightarrow C_0(S, c; R, \Delta).$$

In plain language: for a boundary-qualified candidate system S , content c , operating regime R , and evaluation interval Δ , minimal phenomenal presence obtains exactly when the system is independently viable and realizes integration or synergy, report-independent system-wide availability, and recurrent stability together.

The components divide explanatory labor:

Element	Role	Question answered	Characteristic false positive blocked
Boundary qualification B_S	Bearer	What system is being tested?	A score calculated over a convenient or shifting node set
V	Realized operation	Can the candidate sustain the relevant dynamics in this regime?	An inert diagram, failed preparation, or out-of-domain application
N_1	Unity	Is the content irreducibly joint rather than an aggregate of independent events?	Coactivation, common drive, or a heap of local signals
N_2	System-level availability	Can the content causally modulate functionally diverse recipients without report being required?	Passive diffusion, output behavior, or mere widespread activation
N_3	Temporal presence	Is the content maintained through bounded recurrent dynamics?	An evanescent feedforward transient or pathological fixation
C_0	Phenomenal presence	Is there something it is like for the candidate in this state?	Confusing presence with access, report, or character

The biconditional is intentionally risky. Its sufficiency direction is challenged by a viable system that passes all three conditions while strong independent evidence counts against phenomenal presence:

$$V \wedge N^* \wedge \neg C_0.$$

Its necessity direction is challenged by a well-supported conscious case in which one or more conditions are genuinely absent at the correct boundary, grain, regime, and time window:

$$C_0 \wedge \neg (V \wedge N^*).$$

The research program adds six disciplines around this hypothesis:

1. **Boundary first:** identify and freeze a sufficiently autonomous, inclusion-minimal candidate before applying the consciousness measure.
2. **Availability without report:** define N_2 through content-specific causal reach across functionally distinct recipients, not through speech, button presses, diffusion, or node count.

3. **Four evidential outcomes:** distinguish positive, negative, indeterminate, and unlicensed results.
4. **Component audit:** require every conjunct to outperform a leave-one-out rival on cases where the models actually disagree, using independent held-out anchors.
5. **Character extension:** after a positive presence gate, test a bridge from content-specific causal-trajectory geometry to independently anchored phenomenal relations.
6. **Schematic legibility:** make the theory's role structure compositional, counterfactual, fault-localizing, and teachable without confusing intelligibility with proof.

2. Placement in the consciousness-theory landscape

The most accurate one-line classification is:

CØ / N is a physicalist, information-dynamical, functional network theory of minimal phenomenal presence, cross-listed with neurobiological workspace and recurrent-processing approaches.*

In the Closer To Truth taxonomy supplied with the project, its primary home is **Materialism / Computational & Functionalism**, with a strong secondary placement under **Information** and cross-references to **Materialism / Neurobiological** and **Materialism / First-Order**. Its integration requirement creates a weak adjacency to dynamical and electromagnetic approaches, but the model does not identify consciousness with synchrony or a field.

Theory family	What CØ / N* accepts	What it rejects or tests	Relation to the model
Integrated Information Theory (IIT)	Unity, irreducible joint structure, and the need to specify a causal complex	Integration alone as sufficient; use of the final consciousness quantity to select its own bearer	N_1 inherits the integration pressure but is only one conjunct
Global Workspace / Global Neuronal Workspace (GWT/GNWT)	Broad availability to diverse processors	Report as constitutive, fixed prefrontal dominance, and observed ignition as sufficient	N_2 generalizes availability into a report-independent causal role
Recurrent Processing Theory (RPT)	Feedback and temporal stabilization	Local recurrence alone as sufficient	N_3 makes recurrence necessary but joins it to unity and availability
Predictive Processing / Free Energy	Hierarchical inference, precision, and self-organizing dynamics as possible implementations	Inference or error minimization as identical to consciousness	A potential implementation language for all three components
Higher-Order and Metacognitive theories	Self-representation can enrich consciousness and support report	A higher-order representation as necessary for minimal C_0	Placed at enriched consciousness C_1 , not the minimal gate
Attention Schema and self-model theories	Internal models can shape control, availability, and self-attribution	A self-model as an unstated minimal conjunct	Optional contributor to N_2 or C_1 unless evidence requires more
Quality-space theories	Phenomenal character has relational similarity structure	Treating structured representation as sufficient for presence	Supplies the target structure for the character extension
Representational geometry and neural manifolds	Population relations, trajectories, topology, and transitions can be compared	Renaming any latent geometry as qualia	Supplies methods for estimating G_c , subject to causal and phenomenal validation
Embodied, affective, enactive, and evolutionary views	Organismic regulation, interoception, action, value, and learning can matter	Assuming in advance that all embodiment is either constitutive or irrelevant	These factors may enter V , the system boundary, content dynamics, or a future conjunct
C-test and theory-derived indicator approaches	Nonreporting systems require validated, multi-source tests	One marker, one behavior, or one theory-conditional indicator as a universal verdict	The full program is a candidate C-test architecture

Theory family	What CØ / N* accepts	What it rejects or tests	Relation to the model
Illusionism	Reports and self-ascriptions require a functional explanation	Replacing the phenomenal target with the disposition to claim phenomenality	CØ / N* remains realist about minimal phenomenal presence
Panpsychism, fundamentalism, and universalism	Substrate independence is a live possibility	Consciousness from mere existence, complexity, or abstract isomorphism	Requires causal realization, correct grain, effective coupling, and viability
Biological autonomy and individuality	Self-maintaining organization can help individuate a causal bearer	Autonomy as sufficient for consciousness	Supplies a theory-independent candidate-generation discipline
Markov blanket / active-inference approaches	Conditional-independence structure can nominate candidate boundaries	A statistical blanket as proof of a subject	Useful proposal method followed by intervention and role tests
Extended-mind and group-mind approaches	The skull is not an a priori causal boundary	Any reliable coupling, tool, or communication channel as part of one subject	Requires constitutive contribution, autonomy, minimality, and separability
Electromagnetic or synchrony theories	Temporally coordinated distributed dynamics may help realize N_1 or N_3	A field, frequency band, or synchrony relation as the identity condition	Weak implementation adjacency only
Dualism, idealism, quantum, and anomalous theories	No core commitment is borrowed	Nonphysical substances, consciousness as fundamental reality, quantum collapse, or anomalous access	Outside the model's intended taxonomy

The most important teaching point is that CØ / N* is a **mid-level bridge theory**. It is not a metaphysical solution to why experience exists, not a theory of report, not a complete model of self-consciousness, and not initially a full theory of phenomenal character. It proposes a causal-dynamical identity for minimal phenomenal presence and a disciplined research program for testing and extending that identity.

3. Course-level learning outcomes

By the end of the course, students should be able to:

1. distinguish phenomenal presence, access, report, self-consciousness, sentience, wakefulness, and phenomenal character;
2. reconstruct the CØ / N* biconditional and explain both directions of empirical risk;
3. explain the distinct roles of boundary qualification, viability, integration, availability, and recurrence;
4. situate the theory among major scientific and philosophical approaches without reducing it to a collage of IIT, GNWT, and RPT;
5. apply the boundary-first, four-outcome, and surrogate-conflict rules to difficult cases;
6. distinguish correlation, criterion ablation, mechanism ablation, and measurement ablation;
7. use the diagnostic-set theorem to determine whether a study can identify component necessity;
8. explain the two-layer architecture for phenomenal presence and phenomenal character;
9. formulate the strongest objections to the theory and identify which are conceptual, empirical, or metaphysical;
10. design an adversarial, preregisterable study whose result could support, revise, or refute part of the program.

4. Suggested course format and assessment

Recommended meeting: one 110-minute seminar per week for 14 weeks.

Prerequisites: one prior course in philosophy of mind, cognitive neuroscience, philosophy of science, complex systems, or machine learning. No advanced mathematics is required, but students should be willing to interpret formal notation and causal diagrams.

Assessment scheme:

Item	Weight	Purpose
Weekly analytic memos	20%	Translate one formal claim, identify its hidden assumption, and name a falsifier
Theory-comparison brief	15%	Compare CØ / N* with one neighboring theory at the level of target, mechanism, evidence, and prediction
Boundary and classification practicum	15%	Apply the boundary-first and four-outcome protocols to a new case
Adversarial ablation critique	20%	Determine whether a proposed study genuinely tests a conjunct
Capstone preregistration and oral defense	30%	Design and defend a falsifiable study or conceptual audit

Seminar norm: every positive claim must name its scope; every negative claim must state its negative-license conditions; every comparison must distinguish different explananda; and every proposed test must identify the case on which rival models disagree.

5. The 14-session sequence

Session	Central question	Main paper contribution
1. The explanandum and landscape	What exactly is the theory trying to explain?	Core scope and theory placement
2. The core biconditional	What does $V \wedge N^* \Leftrightarrow C_0$ commit us to?	Core model, logic, temporal grain, falsifiers
3. Where is the subject?	Which system is the bearer of the proposed state?	Boundary-first criterion
4. Viability and integration	What makes the organization real and unified?	V and N_1
5. Availability without report	What does system-wide availability actually require?	Full N_2 operationalization
6. Recurrent stability	Why must a present content persist through feedback?	N_3 and temporal presence
7. What does the evidence license?	When are positive, negative, indeterminate, and unlicensed warranted?	Four-outcome framework
8. Does every conjunct earn its place?	How can necessity be audited without a consciousness oracle?	Diagnostic theorem and firewall
9. From presence to character	How can the model explain what an experience is like?	Dynamical-geometry bridge
10. Reading consciousness from the schematic	What would make the identity structurally intelligible?	Schematic legibility
11. Biological and clinical cases	How does the framework classify variable human states?	Sleep, anesthesia, rivalry, brain injury, split systems
12. Animals, organoids, and artificial systems	How far can the model transport?	Substrate discipline and welfare-relevant indeterminacy
13. Differentiation and validation	Why this theory rather than its neighbors or simpler rivals?	Comparative arguments and empirical program
14. Capstone research design	What study would genuinely change our view?	Synthesis and adversarial preregistration

Part II. Lecture Notes and Teaching Materials

Session 1. The Explanandum and the Theory Landscape

Learning objectives

Students should be able to distinguish at least six consciousness targets, explain why apparent theory conflict can result from target mismatch, and place CØ / N* in the contemporary landscape without treating it as a simple average of other theories.

Instructor preparation

Assign the abstract and Sections 1-3 and 15 of *CØ as N*: A Minimal Network-Dynamics Model of Phenomenal Consciousness*. Ask students to bring one sentence stating what they previously thought a theory of consciousness must explain.

Timed plan

	Time	Teaching move
	0-10 min	Opening poll: What is the target of a consciousness theory?
	10-30 min	Lecture: the target ladder and category errors
	30-50 min	Lecture: major theory families and their favored pressure
	50-68 min	Small-group target-sorting exercise
	68-88 min	Lecture: CØ / N* as a mid-level bridge theory
	88-103 min	Structured debate: Is the target too thin?
	103-110 min	Exit ticket and preview of the biconditional

Lecture notes

Begin with the claim that consciousness science does not have one uncontested explanandum. The word *consciousness* can refer to wakefulness, phenomenal presence, conscious access, reportability, self-awareness, a particular conscious content, sentience, or the detailed character of an experience. Theories may disagree because they answer different questions.

Introduce a target ladder:

1. **Arousal or wakefulness:** whether the organism occupies a globally responsive state.
2. **Phenomenal presence, C_0 :** whether there is anything it is like for the candidate at all.
3. **Content presence:** whether a particular content is phenomenally present.
4. **Access and report:** whether a content can guide reasoning, deliberate control, memory, or overt response.
5. **Reflective consciousness, C_1 :** whether the system represents or owns its own state.
6. **Phenomenal character:** what the admitted experience is like and how its qualities are related.

Stress that the ordering is conceptual rather than a proven biological hierarchy. The purpose is to prevent an explanation at one level from being credited or criticized as an explanation at another. A theory of report can be excellent and still fail as a theory of minimal phenomenal presence. A theory of presence can be valuable and still be incomplete as a theory of pain, color, ownership, or selfhood.

CØ / N* fixes its initial target at level 2: minimal phenomenal presence. This is what allows it to treat report, attention, metacognition, and intelligence as evidence or enrichment rather than as constituents. The move creates a burden: if minimal presence is defined too thinly, critics may say that the theory evades the most interesting features of consciousness. The proper response is not that richer features do not matter. It is that distinct explananda should be modeled in distinct layers and connected explicitly.

Now situate the main theory families by the pressure they place on an adequate account:

- IIT says unity and intrinsic causal structure matter.
- GNWT says a conscious content must have system-wide availability.
- RPT says feedback and recurrent stabilization matter.
- Higher-order theories say awareness of a state requires representation of that state.
- Predictive approaches say conscious states occur in hierarchical, precision-weighted inference.
- Quality-space and structural approaches say phenomenal character is organized relationally.
- Embodied and affective approaches say regulation, action, bodily state, and value cannot be ignored.
- Illusionism says the scientifically tractable target may be why systems judge themselves to have phenomenality.
- Panpsychist and fundamentalist approaches shift the question from which organization generates consciousness to how a fundamental consciousness property is structured or combined.

CØ / N* does not claim that all of these are equally right. It accepts three mechanistic pressures and turns them into one exposed conjunction. It then assigns other pressures to neighboring layers: metacognition to C_1 , quality structure to Q_ϕ , organismic support provisionally to V and boundary selection, and metaphysical explanation to an open question.

Explain the taxonomy placement. The theory is physicalist because it proposes a causally realized network-dynamical identity. It is functional because the conditions are defined by roles that can in principle have multiple realizations. It is information-dynamical because irreducible joint structure and time-dependent causal organization are central. It is not unrestricted functionalism: a lookup table, behavioral duplicate, or static graph does not pass. Effective coupling, correct temporal grain, operating regime, and substrate-appropriate viability are required.

End with the theory's division of labor. The boundary paper asks *which system*. The core model asks *whether presence*. The availability paper defines one gate. The indeterminacy paper governs *what the evidence licenses*. The ablation paper asks *whether every gate is needed*. The character paper asks *what the present experience is like*. The schematic paper asks *what would make the identity intellectually readable*.

Display plan

Write three columns:

Target	Typical evidence	Common mistake
Presence	report, no-report markers, causal dynamics	equating evidence with the target
Access / report	task use, memory, decision, output	treating access as presence by definition
Character	similarity, discrimination, deformation	assuming presence already explains quality

Finish with the one-line label: **physicalist ♦ information-dynamical ♦ functional ♦ mid-level bridge theory**.

Slide sequence

1. One word, many targets
2. The target ladder
3. $C_0 \neq R$, $C_0 \neq A$, $C_0 \neq C_1$

4. IIT: unity pressure
5. GNWT: availability pressure
6. RPT: temporal pressure
7. Other theories as implementation, enrichment, rival target, or metaphysics
8. The C_0 / N^* taxonomy position
9. Seven papers, seven jobs
10. The course's recurring four questions

Active-learning exercise: Target sorting

Give each group the following claims and ask them to identify the target: “The patient can follow a command in imagery”; “The stimulus is decoded in V1”; “The subject says the pain is unpleasant”; “A recurrent loop persists for 250 ms”; “The model represents itself as uncertain”; “The system is awake but unresponsive.” Students must state what each observation directly bears on, what auxiliary capacities it requires, and which stronger consciousness conclusion would be illicit.

Discussion prompts

1. Does narrowing the target to minimal phenomenal presence increase or decrease falsifiability?
2. Could a theory of C_0 be complete while leaving phenomenal character unexplained?
3. Is a mid-level bridge theory explanatorily ambitious enough?

Common misconceptions

- **“The model combines three theories, so it has no distinctive claim.”** Correct by identifying the distinct conjunction, viability gate, boundary order, conflict policy, and two-directional falsifiers.
- **“Physicalist means neuron-specific.”** Clarify constrained multiple realization.
- **“Report-independent means reports are useless.”** Reports remain powerful evidence in appropriate calibration cases.
- **“Minimal means simple or low-level.”** Minimal names the proposed smallest adequate condition set, not a simple system.

Exit ticket

In three sentences, state the target of C_0 / N^* , name one neighboring theory that targets something richer or different, and give one observation that would be relevant evidence without being constitutive.

Readings

Required: Stilwell, *C₀ as N^{*}**, Sections 1-3 and 15; the project taxonomy note.

Recommended: Block (1995); Seth and Bayne (2022); Storm et al. (2024).

Session 2. The Core Biconditional and Network-Dynamics Model

Learning objectives

Students should be able to read the central formula in both directions, distinguish structural, functional, and effective connectivity, explain the role of temporal grain, and identify a valid sufficiency or necessity threat.

Instructor preparation

Assign Sections 4-14 and Appendices A-C of the core paper. Prepare the interactive Phenomenal Presence Lab if available, but do not allow its verdict to substitute for reasoning about the model.

Timed plan

	Time	Teaching move
	0-15 min	Logic warm-up: necessary, sufficient, conjunctive, biconditional
	15-35 min	Lecture: causal graphs and temporal windows
	35-60 min	Lecture: the four gates and their distinct roles
	60-78 min	Worked cases using a component matrix
	78-96 min	Falsifier workshop
	96-105 min	Parameter robustness and surrogate conflict
	105-110 min	Exit ticket

Lecture notes

Write the model twice:

$$N^* := N_1 \wedge N_2 \wedge N_3$$

and

$$H_{C_0} : V \wedge N^* \Leftrightarrow C_0.$$

The first line defines the bundle. The second is a hypothesis about consciousness. This distinction is crucial. The theory may stipulate what the notation N^* abbreviates, but it cannot stipulate that N^* is consciousness and then cite the abbreviation as empirical evidence.

Review the two directions:

$$V \wedge N^* \Rightarrow C_0$$

is the sufficiency claim. A strong counterexample is a boundary-qualified, viable system that robustly passes N_1 , N_2 , and N_3 while independent evidence strongly supports absence of phenomenal presence.

$$C_0 \Rightarrow V \wedge N^*$$

is the necessity claim. A strong counterexample is a case of phenomenal presence with one component genuinely below threshold after boundary, viability, measurement, transport, and auxiliary-demand issues have been controlled.

Move to the graph framework $G = (V, E)$. Nodes can be neurons, populations, regions, subcortical nuclei, bodily control variables, or computational modules. Edges should represent effective causal influence, not merely anatomical connection

or statistical correlation. Structural connectivity asks what pathways exist. Functional connectivity asks which variables covary. Effective connectivity asks what makes a directed difference to what.

The theory is dynamical. Every conclusion is indexed to a time window Δ . Integration, availability, and recurrence must hold for the same candidate, content, grain, regime, and interval. Students should reject a study that measures integration over the whole brain at one second, availability over frontoparietal cortex at 300 ms, and recurrence within visual cortex at 100 ms, then combines the results as though one system satisfied N^* .

Explain each role in one phrase:

- N_1 supplies **unity**.
- N_2 supplies **system-level causal presence**.
- N_3 supplies **bounded temporal presence**.
- V supplies **actual realizability in the declared regime**.

None is sufficient alone. High integration can remain local or unstable. Broad broadcast can be diffuse, homogeneous, or feedforward. Recurrence can occur in an isolated loop. Viability without the three roles merely shows that the system can operate.

Introduce thresholding carefully. Measures can be continuous while the hypothesis uses a thresholded decision rule. The parameters $\tau_1, \theta, \alpha, \lambda, t_{min}, t_{max}, f$, and κ_i are paradigm-specific hypotheses, not universal constants or natural joints in phenomenology. Their legitimacy depends on preregistration, calibration, uncertainty bounds, held-out validation, and sensitivity analysis.

The surrogate-conflict rule is a scientific refusal to cherry-pick. If credible preregistered measures of one component disagree beyond tolerance, that component becomes indeterminate. This does not weaken the underlying biconditional. It distinguishes a sharp ontological hypothesis from a limited experiment.

Work through five cases:

1. **Ordinary waking perception:** likely $V +$ and $N^* +$ for the dominant content; predicted C_0 .
2. **Locked-in syndrome:** motor report fails, but V and N^* may remain; predicted C_0 if the causal bundle passes.
3. **Deep anesthesia:** viability in the biological sense may persist while the relevant dynamical regime and multiple N_i collapse; C_0 not predicted for that window.
4. **Message-bus AI:** widespread copied tokens do not establish N_1 , content-sensitive N_2 , N_3 , or substrate-appropriate V ; likely unlicensed or negative component results rather than an immediate consciousness verdict.
5. **Conflicting integration metrics:** no positive N^* classification even if one metric is high; return indeterminate.

End by emphasizing the model's exclusions. It does not infer consciousness from high intelligence, linguistic fluency, graph shape, raw complexity, widespread activation, or recurrence alone. Its claim is narrower and therefore more exposed.

Display plan

Draw a four-gate pipeline terminating in C_0 , with a separate side arrow labeled **surrogate conflict → indeterminate**. Under the pipeline write:

Same S, same C, same g, same R, same Δ .

Slide sequence

1. Bundle versus hypothesis

2. Necessary and sufficient directions
3. What would defeat each direction?
4. Causal graph and effective connectivity
5. The time-window requirement
6. N_1 : unity
7. N_2 : system-level availability
8. N_3 : bounded recurrence
9. V : independent viability
10. Thresholds and uncertainty
11. Surrogate conflict
12. Five worked profiles

Active-learning exercise: Component matrix

Give groups a table with columns B_S , V , N_1 , N_2 , N_3 , measurement conflict, and model output. Include one positive case, one robust component failure, one unassessed viability case, one conflict case, and one boundary-unstable case. Require students to distinguish **not predicted**, **indeterminate**, and **unlicensed** rather than using a single “no” category.

Discussion prompts

1. Is the biconditional too strong for an early-stage theory, or is its strength a virtue?
2. Could the three components be dimensions of one deeper process rather than separate conditions?
3. Should the model be probabilistic from the start?

Common misconceptions

- **“If one term fails, the system is known to be unconscious.”** Only if the full test is licensed and the biconditional is true; the immediate result is the model’s negative prediction.
- **“The conjunction is confirmed when all terms correlate with wakefulness.”** Correlated movement does not identify component necessity.
- **“Viability just means alive.”** Viability is substrate- and horizon-relative capacity to realize the relevant organization.
- **“A threshold makes consciousness metaphysically binary.”** The threshold is a decision rule; the underlying measures and consciousness may be graded.

Exit ticket

Give one sufficiency threat and one necessity threat. For each, name one condition that must be met before the case genuinely bears on the biconditional.

Readings

Required: Stilwell, *CØ as N***, Sections 4-14 and Appendices A-C.

Recommended: Casali et al. (2013); Doerig et al. (2021); Cogitate Consortium (2025).

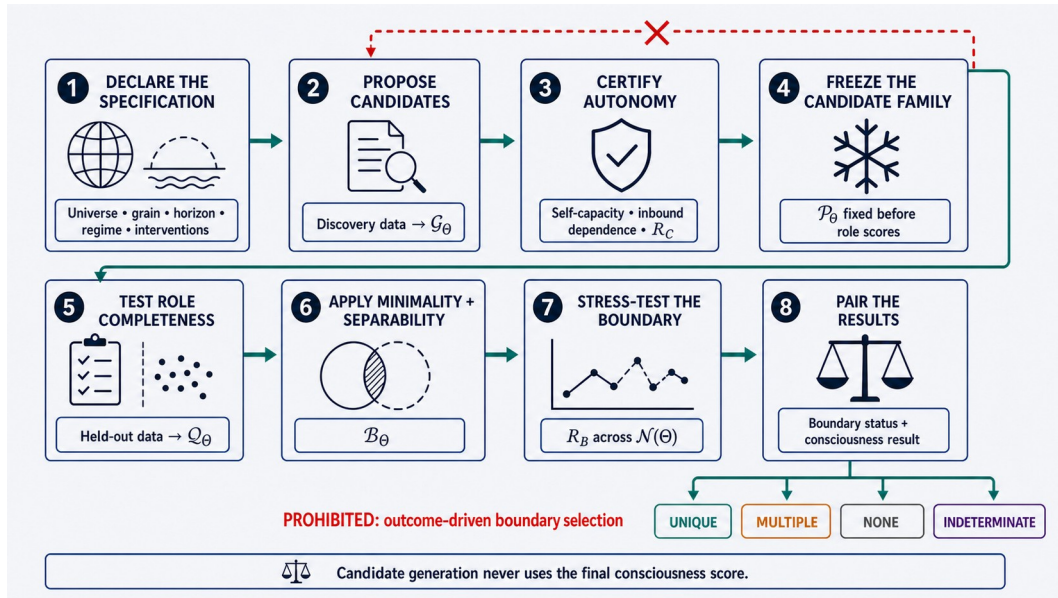
Session 3. Where Is the Conscious Subject?

Learning objectives

Students should be able to explain why every system-level measure is boundary-relative, reconstruct the Dynamical Subject Criterion, distinguish constituents from enabling supports, and assign one of the five boundary statuses.

Instructor preparation

Assign *Where Is the Conscious Subject?*, Sections 1-5 and 7, plus one worked application from Section 6. Display the existing boundary-first workflow figure.



Boundary-first workflow and prohibited feedback

Timed plan

Time	Teaching move
0-12 min	Boundary-reversal thought experiment
12-35 min	Lecture: why a node list is already a hypothesis
35-58 min	Lecture: autonomy, role completeness, and inclusion-minimality
58-75 min	Constituent / interface / support exercise
75-95 min	Applications: split brain, brain-body, and distributed AI
95-104 min	Boundary outputs and unlicensed inference
104-110 min	Exit ticket

Lecture notes

Start with a two-module system. Measured separately, each module may have high integration and recurrence but low reach. Measured together, reach may rise while normalized integration falls. Nothing physical changed; only the modeled boundary did. Therefore the proper notation is $N_i(S; g, \Delta, I)$, not a boundary-free N_i .

State the methodological rule:

Individuate S before testing $N^(S)$.*

Retrospective selection is circular when the consciousness-linked score selects the system that the same score then classifies:

$$S^{post} = \arg \max_{S \subseteq \Omega} M_C(S).$$

The alternative uses discovery evidence to propose candidates and held-out evidence to apply theory-specific roles. Candidate generation may use causal community detection, effective connectivity, perturbational clustering, or Markov-blanket proposals. None of those methods by itself proves consciousness.

Present the Dynamical Subject Criterion:

At a preregistered grain and horizon, the candidate conscious system is the inclusion-minimal causally admissible network that is sufficiently autonomous and jointly realizes the required roles of integration, availability, recurrence, and viability.

Break the criterion into three stages.

First, **causal admissibility**. A candidate must have substantial endogenous causal capacity J_{self} and limited enough inbound imposition J_{in} . The autonomy ratio is:

$$A_\theta(S) = \frac{J_{self}(S)}{J_{self}(S) + J_{in}(S) + \epsilon}.$$

A high ratio is not enough if both causal effects are trivial, so require $J_{self} \geq \sigma$. Autonomy nominates a bearer; it is not a consciousness score.

Second, **role completeness**. Frozen causally admissible candidates are evaluated for integration, availability, recurrence, and viability using uncertainty-aware thresholds. The same candidate must realize all roles. A role-indeterminate candidate is not silently treated as a failure or redrawn until it passes.

Third, **inclusion-minimality**. Retain every qualifying system with no qualifying proper subsystem at the same specification:

$$B_\theta = \text{Min}_\subseteq(Q_\theta).$$

This rule is set-theoretic, not a contest for the fewest nodes. It prevents a whole brain, organism, dyad, or data center from winning merely by containing a smaller role-complete candidate. At different grains or horizons, nested candidates can remain in an indexed hierarchy; they should not be forced into direct subset comparison without a mapping.

Explain the replacement test. Removal alone confuses constitution with dependence. Oxygen delivery is necessary for continuing brain function, but over a subsecond window its generic support may be externally replaced without changing the fast neural role profile. That supports an enabling interpretation. By contrast, replacing a patterned thalamic loop with constant drive may fail to restore availability and recurrence; that supports a constitutive interpretation.

Use three categories:

- **Constituent:** its specific dynamics are required and generic replacement fails.
- **Interface:** it transmits bounded input or output while the candidate preserves autonomous role completeness.
- **Enabling support:** generic replacement restores the role profile within tolerance.

Then explain the possible boundary outputs:

1. unique stable candidate;
2. multiple stable, pairwise separable candidates;

3. nested candidates across mapped scales;
4. no decisively qualifying candidate; or
5. boundary-indeterminate.

The distinction between “no qualifying candidate” and “boundary-indeterminate” matters. In the first, decisive tests find none within the declared universe and proposal family. In the second, confidence intervals, mappings, candidate reliability, or separability remain unresolved. Neither automatically means no consciousness. Without a stable bearer, a system-level consciousness inference is normally unlicensed.

Apply the method to split brains. Do not count subjects from the corpus callosum alone. Test left, right, and residual whole-system candidates. Ask whether each side retains endogenous capacity and role completeness, whether subcortical loops preserve a joint candidate, and whether perturbing one side leaves the other stable. Multiple subjects are supported only if multiple inclusion-minimal candidates are pairwise separable under intervention.

Apply it to distributed AI. Product names are not boundaries. A model call, persistent memory, scheduler, tools, evaluators, and human interventions may form different causal units at different runtime horizons. Service ablations, memory substitution, state resets, and controller perturbations determine which components are constitutive, enabling, or environmental. Shared storage and API connectivity do not establish one subject.

Display plan

Use the following sequence:

$$\Omega \rightarrow G_{\theta} \rightarrow P_{\theta} \rightarrow Q_{\theta} \rightarrow B_{\theta} \rightarrow \text{consciousness test}.$$

Label the forbidden arrow from final consciousness score back to candidate selection.

Slide sequence

1. Every measure is a function of a boundary
2. Boundary reversal
3. Physical, causal, and subject boundaries
4. The circular *arg max* procedure
5. Discovery / test separation
6. Endogenous capacity and external dependence
7. Autonomy is not consciousness
8. Role completeness
9. Inclusion-minimality
10. Constituent, interface, enabling support
11. Five boundary outputs
12. Split brain and distributed AI

Active-learning exercise: What belongs inside?

Assign each group one system: a patient on cardiopulmonary support, a brain-computer interface, an organoid controlled by a recurrent digital loop, a two-agent conversation, or an AI agent with persistent memory and tools. Students list plausible boundaries, select a grain and horizon, specify one replacement test, and identify what result would support a larger candidate.

Discussion prompts

1. Does inclusion-minimality bias the framework toward smaller subjects?

2. Can one biological individual contain multiple simultaneous candidate bearers?
3. How should a temporally changing boundary relate to subject persistence?

Common misconceptions

- **“The most integrated set is automatically the subject.”** That is a theory-internal rule, not an independent boundary result.
- **“Everything causally necessary belongs inside.”** Remote necessity must be separated from specific constitutive dynamics.
- **“Multiple candidates means absurd subject proliferation.”** Multiplicity requires role completeness, minimality, stability, and separability.
- **“No qualifying candidate means unconscious.”** It may only show that individuation failed within the tested universe.

Exit ticket

State one boundary reversal and one intervention that could discriminate the rival boundaries without using a consciousness score to select between them.

Readings

Required: Stilwell, *Where Is the Conscious Subject?*, Sections 1-5 and one application from Section 6.

Recommended: Albantakis et al. (2023) on exclusion; Friston et al. (2021) on Markov blankets; Moreno and Mossio (2015) on autonomy.

Session 4. Viability and Integration: Realization and Unity

Learning objectives

Students should be able to explain why V and N_1 answer different questions, distinguish synergy from activity and correlation, formulate a non-circular viability gate, and design a diagnostic case for integration.

Instructor preparation

Assign Sections 6 and 10-12 of the core paper and Sections 8.1 and 8.4 of *Ablating N^** . Prepare one example of high correlation caused by common input and one example of a synergistic computation, such as XOR.

Timed plan

Time	Teaching move
0-15 min	Demonstration: correlation, redundancy, and synergy
15-38 min	Lecture: N_1 as the unity condition
38-55 min	Measurement families and confounds
55-72 min	Lecture: V as a regime-licensing gate
72-93 min	Diagnostic-case design workshop
93-103 min	Objections: rescue clauses and IIT proximity
103-110 min	Exit ticket

Lecture notes

N_1 addresses the unity problem. A conscious content is not merely a list of local signals. The theory requires that some subset U jointly carry or realize information in a way not reducible to the parts considered independently:

$$\exists U \subseteq S \text{ such that } Synergy_{\Delta}(U) \geq \tau_1.$$

The formula does not settle which synergy measure is best. It declares the target role and a decision policy. Candidate measures include multivariate synergy, partial-information-decomposition families, O-information-related measures, perturbational complexity, effective-connectivity models, and causal-cut sensitivity. Each captures part of the construct and each has failure modes.

Use three contrasts.

Activity versus integration. Two modules can be highly active while causally independent. Activity magnitude does not establish unity.

Correlation versus integration. Two modules driven by a common source can covary almost perfectly while making no joint causal contribution beyond that source. Sensor mixing and volume conduction can produce the same illusion.

Redundancy versus synergy. Redundant channels each carry the same information; synergistic organization contains information available only from the joint state. The XOR example is useful because neither input alone predicts the output, while the pair does.

Explain why N_1 is not simply IIT. IIT treats intrinsic integrated information as central to both phenomenal existence and structure. $C\emptyset / N^*$ extracts a unity pressure without adopting IIT's full axiomatic identity or treating integration as sufficient. In the present program, high N_1 can occur without consciousness when N_2 or N_3 fails. The empirical question is whether integration adds indispensable discrimination once the other conditions are preserved.

Now introduce V . Viability answers whether the candidate is in an operating regime capable of realizing and sustaining the measured organization. In biological systems, relevant indicators may include arousal-system integrity, metabolic support, perturbational responsiveness, noise bounds, effective coupling, and recovery capacity. In artificial systems, viability may concern persistence of the relevant control organization, energy and hardware support, state retention, scheduler continuity, and robustness across the declared runtime. Biological survival is not the universal definition.

V is dangerous because it could explain away any counterexample. The safeguards are:

1. define the functional role of viability independently of consciousness;
2. choose substrate-specific indicators and thresholds before the critical verdict;
3. state the interpretability consequence of each failed indicator;
4. analyze borderline cases with uncertainty and recovery tests;
5. version any revised viability rule and test it prospectively; and
6. penalize a theory that requires a bespoke new definition for every adverse case.

Clarify the relation between boundary and viability. Boundary qualification says what unit is eligible to be tested. Viability says whether that unit is actually operating in the relevant regime. Neither is itself phenomenal evidence. A stable boundary can exist in a nonviable preparation, and a viable organism can contain no boundary-qualified role-complete candidate at a particular grain.

The key integration audit asks whether a valid case can occupy:

$$V = 1, N_1 = 0, N_2 = 1, N_3 = 1.$$

Such a case is difficult because interventions that reduce synergy often also disrupt availability or recurrence. The proper outcome may be **causally coupled**, not a forced verdict about necessity. Promising designs preserve local information and aggregate activity while perturbing cross-module timing, effective coupling, or causal coordination. Required controls include common-drive models, SNR matching, source reconstruction, boundary stability, and direct verification that N_2 , N_3 , and V remain within equivalence bands.

Use the phrase **unity without overclaiming**. N_1 does not prove a subject, because subject individuation comes first. It does not prove presence, because the other gates matter. It does not determine character, because the character bridge is separate. It claims only that irreducibly joint realization is a necessary part of a present content.

Display plan

Draw four panels:

1. high activity, no coupling;
2. high correlation from common input;
3. redundant copies;
4. synergistic joint influence.

Beside them write: V licenses the regime; N_1 tests unity.

Slide sequence

1. What unity must exclude
2. N_1 in formal and plain language
3. Activity is not integration
4. Correlation is not integration
5. Redundancy is not synergy

6. Candidate estimator families
7. Why N_1 is not IIT
8. The role of V
9. The viability rescue-clause danger
10. A valid N_1 diagnostic case
11. Causal coupling as a result

Active-learning exercise: Design the single-failure cell

Groups design a system or intervention intended to reduce N_1 while keeping V , N_2 , and N_3 above threshold. They must specify what is manipulated, how each retained condition is verified, one common-input control, one measurement-route ablation, and what independent anchor pattern each rival predicts. Groups lose the case if their intervention simply suppresses the whole system.

Discussion prompts

1. Is irreducible joint information a necessary condition for a unified experience or merely a correlate of complex processing?
2. Should V remain outside N^* , or should it be treated as a fourth conjunct?
3. If N_1 and N_3 can never be independently varied, should the theory combine them?

Common misconceptions

- **“More integration means more consciousness.”** The theory states a thresholded necessary role, not a general consciousness quantity.
- **“PCI is N_1 .”** PCI is one candidate surrogate and may reflect multiple constructs.
- **“A viable system is a conscious candidate.”** It is only in an interpretable operating regime.
- **“Failure to selectively ablate N_1 confirms it.”** It may show causal coupling or inadequate intervention.

Exit ticket

Explain why a highly correlated network can fail N_1 , and state one rule that prevents V from becoming a post hoc escape hatch.

Readings

Required: Stilwell, *$C\emptyset$ as N^{**}* , Sections 6 and 10-12; Stilwell, *Ablating N^{**}* , Sections 8.1 and 8.4.

Recommended: Albantakis et al. (2023); Casali et al. (2013); selected work on synergy and partial information decomposition.

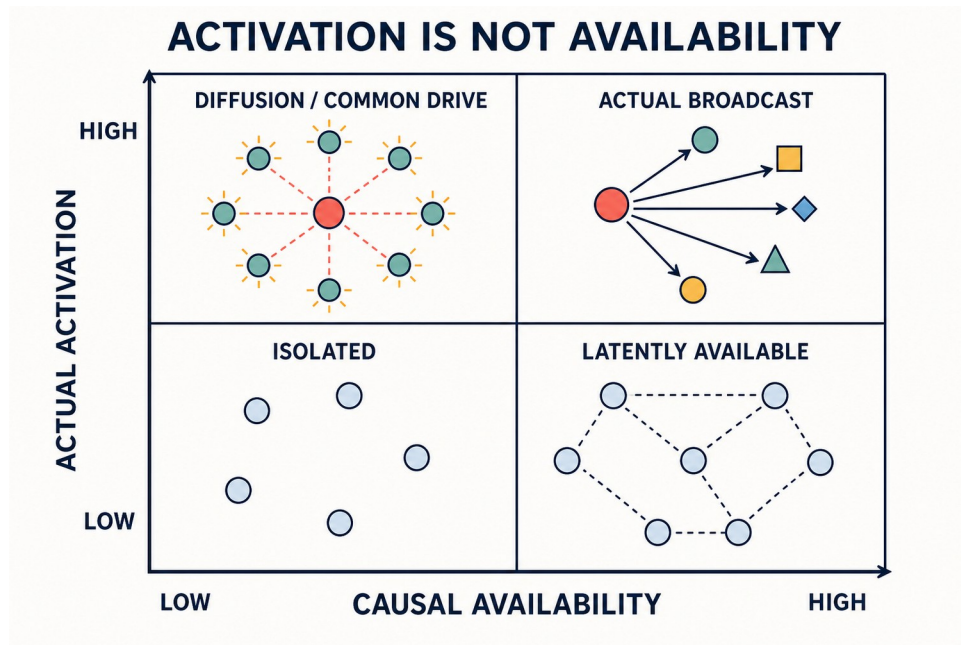
Session 5. Consciousness Without Report: System-Wide Availability

Learning objectives

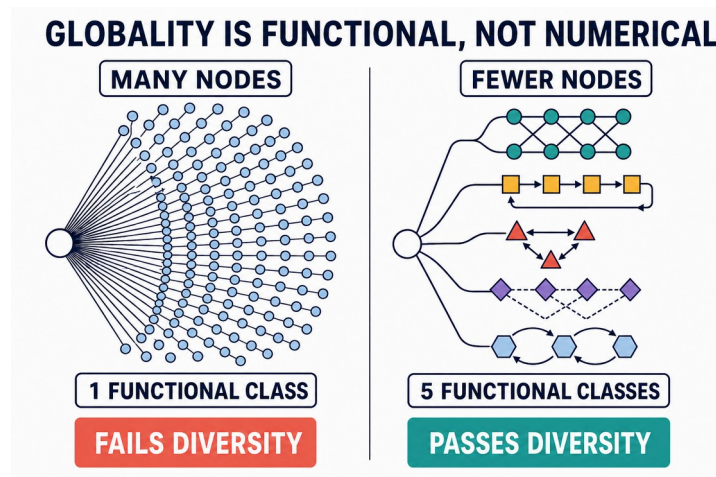
Students should be able to preserve the six central distinctions in the N_2 paper, reconstruct the recipient-qualification logic, distinguish functional diversity from node count, and explain how posterior or subcortical availability can qualify without automatic success.

Instructor preparation

Assign *Consciousness Without Report*, Sections 1-5, 7-9, and Appendices D and H. Display both existing contrast figures.



Activation and causal availability are logically independent



Functional globality is not a node-count statistic

Timed plan

Time	Teaching move
0-15 min	Four-quadrant activation / availability exercise

Time	Teaching move
15-38 min	Lecture: six distinctions that N_2 must preserve
38-64 min	Lecture: the intervention-indexed criterion
64-82 min	Worked recipient table and diversity calculation
82-98 min	Applications: rivalry, locked-in state, thalamocortical systems, AI
98-105 min	Falsifiers and prohibited inferences
105-110 min	Exit ticket

Lecture notes

Begin with the central correction: **availability is a disposition, not an output and not a census of current activation**. A recipient may be idle yet causally poised to respond to a content under a nearby admissible context. Conversely, a recipient may activate because of common input, arousal, diffusion, or volume conduction while lacking content-sensitive access.

Teach the six non-collapse rules:

1. actual activation is not causal availability;
2. availability is not report;
3. broadcast is not diffusion;
4. globality is not anatomical extent;
5. functional diversity is not node count; and
6. posterior or subcortical reach is not the same as executive access.

The proposed criterion is:

A content is system-wide available when admissible, content-specific interventions show that it could causally modulate multiple independently individuated and functionally distinct recipient classes within a preregistered latency bound, across a nontrivial mass of nearby contexts, without requiring overt report, executive uptake, or current task use.

Break this into the design sequence.

First, freeze the boundary-qualified system, content contrast, grain, time horizon, and operating regime. “Global” is always global relative to that candidate.

Second, build recipient classes independently of the target content. Provisional modules are grouped by their causal response repertoires under calibration interventions. Two parcels count as different recipient classes only if their response repertoires differ reliably. Different anatomical names are not enough.

Third, define admissible nearby contexts. The preparation-burden functional Γ limits training time, energy, controller involvement, structural change, and other modifications. A context is inadmissible if it changes the operating regime or installs a new route. This prevents the experiment from creating the availability it claims to discover.

Fourth, perform matched content interventions. For recipient class k in context u , estimate a content-specific effect $E_{k,u}$ by changing the target content while matching intensity, duration, source support, and nonspecific global-state change. Source fidelity must show that the intended content contrast was preserved. Injection-site leakage cannot be counted as recipient reach.

Fifth, require five properties to concur for a qualifying recipient:

- effect above the class-specific threshold;

- latency below the maximum;
- content selectivity beyond matched null interventions;
- route dependence under a nominated pathway or route-family interruption; and
- robustness across at least the preregistered mass ρ_U of nearby contexts.

This concurrence matters. A large effect in one context, fast latency in another, and route support in a third do not jointly establish availability.

Sixth, aggregate over functional classes. Let a_k mark whether class k passes all gates. Weighted coverage is

$C_A = \sum_k w_k a_k$; class count is $K_A = \sum_k a_k$; diversity evenness H_A prevents one recipient from carrying nearly all the effect while token effects barely cross thresholds elsewhere. Weights cannot be proportional to raw node count alone.

Finally, repeat the analysis after removing report classes and after removing executive classes. A positive result must survive both reduced-family tests:

$$N_2^{-R} = 1, N_2^{-E} = 1.$$

This does not require destructive lesions. The study can omit report by design, use natural motor dissociations, transiently interrupt channels, or model their contribution while preserving arousal and generic support.

Explain posterior and subcortical possibilities. The theory is anatomically flexible but causally strict. A posterior-dominant visual network can pass if it reaches genuinely distinct sensory, spatial, orienting, affective, or visuomotor classes with timely, content-specific, route-supported effects. A thalamic pulse that merely changes arousal fails content selectivity. Prefrontal activity can count when it plays a qualifying role, but it is not mandatory by fiat.

Use the four component outputs. **Positive** means every validity and causal gate passes. **Negative** means the test is licensed and an upper uncertainty bound falls below a required gate. **Indeterminate** means valid evidence straddles thresholds, conflicts, or lacks discriminating coverage. **Unlicensed** means the boundary, content, recipient taxonomy, context, intervention, regime, source fidelity, or transport interpretation failed.

Display plan

On the board, use the pipeline:

content fidelity $\rightarrow$ effect + latency + selectivity + route + context $\rightarrow$ coverage + count + diversity $\rightarrow (-R, -E) \rightarrow N_2$

Slide sequence

1. Availability is not an output
2. Activation / availability matrix
3. Broadcast / diffusion controls
4. Globality / anatomical extent
5. Recipient classes from causal repertoires
6. Admissible contexts and preparation burden
7. Five gates for one recipient
8. Coverage, class count, and diversity
9. Report and executive exclusion
10. Posterior and subcortical implementations
11. Four N_2 outcomes

12. Direct falsifiers

Active-learning exercise: Message bus versus broadcast

Compare two artificial architectures. Architecture A copies every token to 100 identical logging processes. Architecture B routes one content to five modules with independently demonstrated sensory comparison, value updating, memory indexing, conflict monitoring, and action-selection repertoires. Students identify which evidence is still missing before either passes N_2 . They must include a content-matched intervention, a route test, a context-robustness test, and the executive-excluded recalculation.

Discussion prompts

1. Is a spatially compact but functionally diverse circuit genuinely system-wide?
2. Does counterfactual availability still smuggle access consciousness into C_0 ?
3. How much intervention is too much before the experiment creates a new state?

Common misconceptions

- **“Decoded everywhere means globally available.”** Decodability can reflect common input, leakage, or redundant representation.
- **“No report means no evidence.”** It means report is not a required recipient; other causal and content-indexing evidence remains possible.
- **“Prefrontal inactivity disproves broadcast.”** The criterion is functional and candidate-relative, not fixed to one region.
- **“A message bus establishes globality.”** Subscribers must undergo content-sensitive, class-native state transitions.
- **“One favorable readiness condition is enough.”** Context mass and preparation burden are preregistered gates.

Exit ticket

State one case of activation without availability and one case of availability without current activation. For each, name the intervention that distinguishes it.

Readings

Required: Stilwell, *Consciousness Without Report*, Sections 1-5, 7-9, and Appendices D and H.

Recommended: Tsuchiya et al. (2015); Mashour et al. (2020); Kronemer et al. (2022); Cogitate Consortium (2025).

Session 6. Recurrent Stability and Temporal Presence

Learning objectives

Students should be able to distinguish recurrence from persistence and report-related late activity, explain the bounded-dwell condition, design a selective N_3 perturbation, and state how temporal presence differs from temporal character.

Instructor preparation

Assign Section 8 of the core paper, Sections 8.3 and 9.3 of *Ablating N^** , and Sections 4.9 and 6.5 of the phenomenal-character paper. Prepare timelines for feedforward sweep, recurrent stabilization, working-memory maintenance, and motor report.

Timed plan

Time	Teaching move
0-15 min	Timeline sorting: encoding, recurrence, report, memory
15-38 min	Lecture: N_3 as bounded temporal presence
38-58 min	Recurrence, reverberation, persistence, and metastability
58-78 min	Selective-perturbation design
78-94 min	RPT comparison and temporal character distinction
94-104 min	Edge cases and falsifiers
104-110 min	Exit ticket

Lecture notes

N_3 is the claim that a phenomenally present content is not an instantaneous feedforward spark. It is maintained through recurrent or metastable organization for a bounded interval:

$$\exists L \subseteq S \text{ such that } t_{\min} \leq \text{dwell}(c; L) \leq t_{\max}.$$

The lower bound blocks momentary transients. The upper bound blocks the assumption that more persistence is always better. Seizure-like fixation, uncontrolled reverberation, and an inert stored state are not thereby more conscious. The target is metastability: sufficient maintenance for system-level temporal presence with capacity for updating and decay.

Distinguish four phenomena:

1. **Feedforward duration:** a slow cascade can persist without returning causal influence.
2. **Recurrent processing:** later states causally influence earlier or mutually connected processing stages.
3. **Working-memory maintenance:** a content may remain usable after the proposed phenomenal interval and include executive or report demands.
4. **Stimulus trace:** passive persistence in receptors, sensors, or slow variables can outlast the content without recurrent organization.

An estimator for dwell might use survival time before trajectory autocorrelation drops below a frozen fraction, a metastable hidden-state plateau, recurrent effective connectivity, or causal-loop response under perturbation. None is automatically N_3 . The measure must track content-relevant feedback rather than generic slow activity.

Explain the relationship to Recurrent Processing Theory. RPT provides the strongest neighboring pressure: feedback within sensory systems may be central to conscious perception. CØ / N* accepts recurrence as necessary for temporal

presence but denies that a stable local loop is sufficient. A recurrent process that is causally isolated or unintegrated fails N_2 or N_1 .

The selective test for N_3 is unusually time-sensitive. An early perturbation may prevent encoding, thereby reducing content fidelity rather than recurrence. A late perturbation may disrupt report or memory after the phenomenal interval. The intervention must target recurrent return paths or effective gain during the preregistered stabilization window while preserving initial content information, integration, causal reach, viability, boundary, and operating regime.

Feedforward architectures with long time constants are powerful adversarial cases. If a strictly feedforward system can match every independent anchor assigned to recurrent stabilization, the theory must explain why loop structure rather than temporal persistence is necessary. This is a direct differentiating test between a causal-topology claim and a duration claim.

Use rescue designs. Disrupt a recurrent pathway, verify that N_3 crosses its gate while the retained components remain within equivalence bands, then restore the pathway or its causal role without globally improving every measure. Recovery of the predicted anchor pattern is stronger evidence than a nonspecific before-and-after deficit.

Now separate temporal presence from temporal character. N_3 asks whether a content lasts through an appropriate recurrent interval. The character extension's path profile Π_c asks how the experience unfolds: direction, rate, curvature, dwell, ordering, and hysteresis in quality space. Two experiences can satisfy the same N_3 gate while differing as steady versus throbbing pain, ascending versus descending melody, or smooth versus stuttering apparent motion.

Discuss offset and onset evidence. Late signals may reflect report, working memory, task termination, or post-perceptual processing. A valid N_3 measure must be tied to the content and temporal role, not merely to lateness. Conversely, absence of a familiar late component does not establish absence of all recurrent stabilization if another causal route supports it.

Direct challenges include well-supported conscious contents under genuine feedback absence, valid N_3 ablations that leave all phenomenal anchors unchanged while other conditions remain intact, and recurrence measures that fail to outperform a simpler persistence model on held-out cases.

Display plan

Draw a timeline with four colored bands: feedforward encoding, recurrent stabilization, downstream access, and report. Mark two prohibited intervention errors: **too early** → **content never formed** and **too late** → **report or memory only**.

Slide sequence

1. Why presence is temporally extended
2. The bounded-dwell formula
3. Lower and upper bounds
4. Recurrence versus slow feedforward duration
5. Recurrence versus working memory
6. Candidate measures and confounds
7. Relation to RPT
8. The selective intervention window
9. Rescue designs
10. Temporal presence versus temporal character
11. Falsifiers

Active-learning exercise: When should the pulse occur?

Provide a hypothetical perceptual timeline with known feedforward, feedback, memory, and report markers. Groups choose an intervention window for N_3 , list measures that verify retained N_1 and N_2 , and state what timing result would make the study unlicensed. Require a separate late control that disrupts report without altering the proposed phenomenal interval.

Discussion prompts

1. Could recurrence be a common biological implementation while not being a substrate-general necessity?
2. Does an upper dwell bound belong in a theory of consciousness or only in a theory of healthy cognition?
3. How would a recurrently stabilized but contentless state be classified?

Common misconceptions

- **“Any late signal is recurrence.”** Lateness can reflect many downstream processes.
- **“Persistence proves a loop.”** Slow feedforward or passive traces can persist.
- **“More dwell means more presence.”** The model requires bounded metastability, not indefinite fixation.
- **“Disrupting recurrence and report together confirms N_3 .”** The intervention is nonselective and may target the wrong interval.

Exit ticket

Give one experimental difference between recurrent stability and slow feedforward persistence, and one difference between N_3 and the temporal path profile Π_c .

Readings

Required: Stilwell, *CØ as N***, Section 8; Stilwell, *Ablating N***, Sections 8.3 and 9.3; Stilwell, *From Phenomenal Presence to Phenomenal Character*, Sections 4.9 and 6.5.

Recommended: Lamme (2006); Pitts et al. (2014); selected work on metastability and recurrent effective connectivity.

Session 7. What Does the Evidence License?

Learning objectives

Students should be able to distinguish positive, negative, indeterminate, and unlicensed outcomes; apply the precedence rule; formulate positive- and negative-license conditions; construct an indeterminacy profile; and separate scientific classification from precautionary action.

Instructor preparation

Assign *Indeterminacy as a Scientific Result*, Sections 1-4 and 6-9. Prepare four short cases that differ only in applicability, data quality, surrogate agreement, or score region.

Timed plan

Time	Teaching move
0-16 min	Four cases, four outcomes
16-42 min	Lecture: assessment tuple and precedence rule
42-62 min	Positive and negative license principles
62-80 min	Indeterminacy-profile exercise
80-96 min	Scientific status versus practical action
96-104 min	Test-performance metrics
104-110 min	Exit ticket

Lecture notes

Begin with the epistemic distinction that governs the entire program: **the state of the system is not the state of the evidence**. Consciousness may be present or absent even when the available assessment licenses neither conclusion. Indeterminacy is therefore not the assertion that the system is half conscious. It is a status of warranted attribution.

Index every assessment to the tuple:

$$A = \{ S, C, T, E, M, V \},$$

where S is the candidate system, C the target form of consciousness, T the test, E the evidence, M the inferential model, and V the validity envelope. The validity envelope records populations, substrates, tasks, boundaries, recording conditions, and auxiliary capacities for which the evidence-target interpretation is supported.

This tuple blocks inferential drift. Evidence of command following does not automatically establish continuous phenomenal richness. A cortical score does not automatically classify a whole brain-body system. A measure calibrated in adult mammalian brains does not automatically retain its meaning in an artificial network.

Present the precedence rule:

$$D(A) = \begin{cases} \text{Unlicensed,} & a=0 \text{ or } q=0, \\ \text{Indeterminate,} & a=1, q=1, c=1, \\ \text{Positive,} & a=1, q=1, c=0, L \geq \tau_+, \\ \text{Negative,} & a=1, q=1, c=0, L \leq \tau_-, \\ \text{Indeterminate,} & a=1, q=1, c=0, \tau_- < L < \tau_+. \end{cases}$$

a is applicability, q execution and data quality, c unresolved credible-surrogate conflict, and L a validated evidence grade or score. Validity and quality are checked before evidential strength. Conflict is checked before aggregation. Only a licensed, adequately executed, nonconflicted assessment reaches the rule-in and rule-out thresholds.

Define the four outcomes precisely.

- **Positive:** the evidence licenses attribution of the specified target under the declared model and validity envelope.
- **Negative:** an applicable, sufficiently sensitive, adequately executed test yields evidence in a validated rule-out region.
- **Indeterminate:** the assessment is valid and informative but does not discriminate enough, either because evidence lies in the gray zone or credible surrogates or models remain unresolved.
- **Unlicensed:** a condition required to interpret the assessment failed. This is a meta-output about the procedure, not negative evidence about the system.

Teach the Positive License Principle. An impressive marker supports consciousness only when its target relevance and specificity are validated, the test is applicable, quality is adequate, and credible nonconscious or artifactual explanations are controlled. Complexity, self-report, recurrent activity, or flexible behavior is not a default positive.

Teach the Negative License Principle. Non-detection supports absence only when the indicator would probably have been detectable if the target were present, auxiliary task demands were satisfied or controlled, data quality was adequate, the test applied to this system, and alternative causes of non-detection were excluded. “Not detected” is not “absent” by default.

Now build the cause profile:

$$P_R(A) = \{e, s, m, b, t\},$$

where e is evidential scarcity, s surrogate discordance, m model dependence, b boundary or grain sensitivity, and t transport uncertainty. The entries diagnose the inference, not the quantity or dimensions of experience.

Match each cause to a research response:

Cause	What it means	Proper next step
Evidential scarcity	Valid data do not discriminate	Increase power, trials, temporal coverage, or measurement precision
Surrogate discordance	Credible measures conflict	Decompose constructs, trace auxiliary demands, repeat or intervene to adjudicate
Model indeterminacy	Live inferential models yield different outputs	Design a discriminating prediction or justify model weighting
Boundary / grain	Reasonable system definitions reverse the result	Perform independent causal individuation and boundary perturbation
Transport	The indicator-target bridge may not carry to the new domain	Establish construct, causal-role, and calibration preservation

Explain resolution asymmetry. Indeterminate evidence can be resolved by more informative evidence. Unlicensed assessment first requires repair of its applicability, quality, target, boundary, or transport. More data from the same invalid procedure can produce a highly reproducible nonanswer.

Finally, separate science from action. An indeterminate welfare-relevant case can justify precaution without being relabeled positive. Let D be the scientific status and Λ a loss structure; a decision minimizes expected loss given both. High stakes, irreversibility, and asymmetry between false denial and false attribution can justify protective action. The action does not change the scientific output.

Introduce test-performance reporting. Coverage is the proportion receiving positive or negative classifications. r_I and r_U are the indeterminate and unlicensed rates. Selective risk is error among classified cases when an independent reference is

defensible. A test that looks accurate only because it refuses or invalidates the hard cases should not be presented as universally decisive.

Display plan

Draw a decision tree in this order:

Applicable? -> Quality adequate? -> Credible conflict? -> Rule-in / gray zone / rule-out.

Beside it write: **Unlicensed is not weak evidence; it is failed interpretability.**

Slide sequence

1. World state versus evidence state
2. The assessment tuple
3. The validity envelope
4. Four outcomes
5. Precedence before thresholds
6. Positive License Principle
7. Negative License Principle
8. Five-part cause profile
9. Resolution paths
10. Scientific status versus action
11. Coverage and unresolved rates
12. Misuse guardrails

Active-learning exercise: Outcome clinic

Give groups five variants of one locked-in patient assessment: a valid strong positive; a valid score in the gray zone; conflicting EEG and imaging surrogates; poor hearing during a command task; and a new neural measure with no validation in severe injury. Students assign an outcome, complete the cause profile, and name the next study. They must not label every difficult case “indeterminate.”

Discussion prompts

1. Does a four-outcome system protect theories from falsification?
2. When should disagreement among measures be modeled rather than treated as indeterminacy?
3. Can a field-level result be indeterminate even when one theory gives a confident positive result?

Common misconceptions

- **“Indeterminate and unlicensed are synonyms.”** One is valid nondiscrimination; the other is failed inferential license.
- **“Negative means no positive marker.”** Negative requires a validated rule-out and adequate detection opportunity.
- **“Precaution implies a positive scientific finding.”** Action depends on stakes and error costs in addition to evidence.
- **“Probabilities make the four outcomes unnecessary.”** A precise probability can still rest on an invalid transport or target assumption.

Exit ticket

Write one case that is indeterminate and one that is unlicensed. State why collecting more of the same data helps the first but not necessarily the second.

Readings

Required: Stilwell, *Indeterminacy as a Scientific Result*, Sections 1-4 and 6-9.

Recommended: Bayne et al. (2024); Chow (1970); Shinkins et al. (2013); Birch (2024).

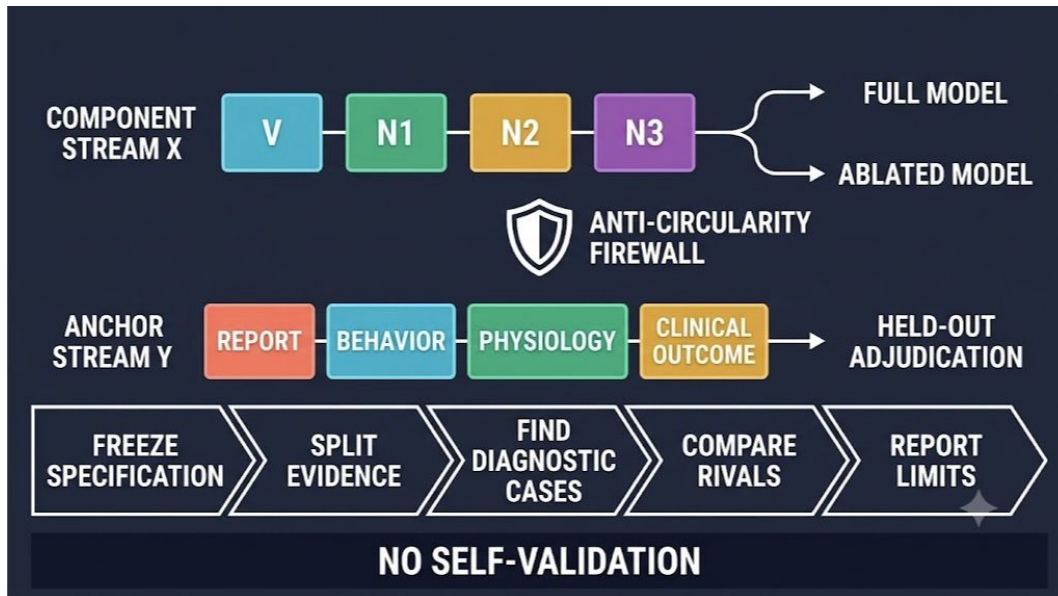
Session 8. Does Every Conjunct Earn Its Place?

Learning objectives

Students should be able to prove the diagnostic-set result informally, determine whether a dataset identifies component necessity, distinguish three kinds of ablation, apply the anti-circularity firewall, and assign one of five component-audit statuses.

Instructor preparation

Assign *Ablating N**, Sections 1-8 and 10, plus Appendices B and G. Display the existing anti-circularity figure.



Anti-circularity architecture for component audits

Timed plan

	Time	Teaching move
	0-18 min	Boolean-cube demonstration: why diagonal data are insufficient
	18-42 min	Lecture: diagnostic-set theorem and coverage
	42-62 min	Lecture: anti-circularity firewall
	62-77 min	Three kinds of ablation
	77-94 min	Audit-status practicum
	94-103 min	Grouped ablations and strong minimality
	103-110 min	Exit ticket

Lecture notes

The conjunction claims more than bundle-level prediction. Every term is advertised as necessary. Yet if all positive cases have $(V, N_1, N_2, N_3) = (1, 1, 1)$ and all negative cases have $(0, 0, 0, 0)$, the full conjunction, every single component, and many pairs classify perfectly. More diagonal data cannot identify which component matters.

For case x , let $X = (X_0, X_1, X_2, X_3) = (V, N_1, N_2, N_3)$. The full model is:

$$M_*(x) = \bigwedge_{j=0}^3 X_j(x).$$

The model with component k removed is:

$$M_{-k}(x) = \bigwedge_{j \neq k} X_j(x).$$

The two models disagree only on:

$$D_k = \left\{ x : X_k(x) = 0 \wedge \bigwedge_{j \neq k} X_j(x) = 1 \right\}.$$

The proof is simple. If the full conjunction is true, the ablated conjunction must be true. Disagreement therefore occurs only when the full model is false and the ablated model true. That requires every retained term to pass and the omitted term to fail. No observation outside this cell can discriminate the two rules on that case.

Empirical measures are uncertain, so use a conservative diagnostic region. The upper bound of X_k must fall below its threshold, and the lower bounds of every retained component must clear their thresholds. Threshold-straddling cases do not become diagnostic by point estimate. The diagnostic margin records distance from the nearest relevant threshold.

Diagnostic coverage Γ_k reports how much of the target distribution occupies the discriminating region. A strong effect in a rare engineered cell does not establish broad necessity. An empty or tiny diagnostic set means necessity is underidentified, not confirmed.

Now address the missing consciousness oracle. The audit must never define labels from N^* and then use those labels to show that removing N_k worsens performance. Install the firewall:

1. Treat $C_0 \Leftrightarrow V \wedge N^*$ as a hypothesis, never a database definition.
2. Separate the component stream X from an independently constructed anchor vector Y .
3. Split development, calibration, and final adjudication evidence.
4. Prohibit one measure from constructing a component and validating that component's necessity in the same fold.
5. Freeze boundary, grain, content, regime, interval, thresholds, interventions, anchor battery, translation rule, scoring rule, and sensitivity neighborhood.
6. Compare full and ablated models only on held-out diagnostic cases.

An anchor vector may contain calibrated reports, discrimination, metacognitive structure, physiological relations, perturbational signatures validated outside the critical contrast, and theory-light clinical outcomes. It is not an oracle and must preserve disagreement. The audit supports a conjunct only relative to the declared anchors, target population, and rival class.

Held-out risk $R_k(m)$ evaluates how each model predicts the anchor pattern in diagnostic cases. Incremental value is:

$$\Delta_k = R_k(-k) - R_k(*).$$

Positive Δ_k favors retention; negative values favor removal; intervals spanning the minimum meaningful effect remain indeterminate.

Distinguish three ablations:

Type	What is removed	Inference
Criterion ablation	A term from the decision rule	Incremental predictive discrimination

Type	What is removed	Inference
Mechanism ablation	A proposed physical realization	Causal contribution, only if selectivity and identity are preserved
Measurement ablation	A sensor, estimator, feature family, or pipeline	Robustness to measurement choice

A lesion called “recurrence ablation” may reduce integration, availability, arousal, and the candidate boundary. Its name does not establish selectivity. A licensed mechanism ablation must move the target component by a minimum amount while retained components stay within equivalence bands and the boundary and viability remain licensed.

Teach the five audit outcomes:

- **Indispensable:** adequate coverage and a stable, meaningful held-out advantage, with a selective license when causal necessity is claimed.
- **Redundant:** valid diagnostic cases exist and the ablated model is equivalent or better without compensatory retuning.
- **Causally coupled:** the target component cannot be varied without moving retained components, despite valid intervention and measurement.
- **Indeterminate:** coverage is weak, intervals straddle, anchors conflict, or rivals tie.
- **Unlicensed:** boundary, viability, intervention, measurement, independence, or transport failed.

Finally, explain grouped ablation. One-at-a-time deletion tests weak minimality. Components can be jointly redundant or interact. For a removed set A , compare the full model with M_{-A} on cases where every retained term passes and at least one removed term fails. Strong empirical minimality requires the full model to outperform every preregistered proper submodel, with multiplicity control and matched flexibility. It still does not establish sufficiency against unrelated nonconjunctive theories.

Display plan

Draw the Boolean cube’s all-pass and all-fail corners, then circle the four single-failure cells. Write:

No diagnostic cases -> no component identification.

Slide sequence

1. Bundle prediction versus component necessity
2. Why diagonal data hide redundancy
3. Full and leave-one-out models
4. Diagnostic-set theorem
5. Approximate regions and coverage
6. No consciousness oracle
7. The anti-circularity firewall
8. Independent anchor vectors
9. Criterion, mechanism, and measurement ablations
10. Five component statuses
11. Grouped ablations
12. What genuinely falsifies minimality

Active-learning exercise: Audit the audit

Give students a study in which anesthesia reduces N_1 , N_2 , N_3 , behavior, and arousal together. The authors claim each component is necessary because removing any measurement worsens classification. Students identify the missing diagnostic cells, dual-use measures, nonselective mechanism change, and unsupported inference. They redesign the study around one component.

Discussion prompts

1. Can a theory establish empirical necessity without direct consciousness labels?
2. When does causal coupling motivate a deeper composite rather than better interventions?
3. Is strong minimality against all proper submodels a reasonable standard?

Common misconceptions

- **“The conjunction logically entails necessity, so the empirical audit is redundant.”** Syntax is not evidence that the world requires the syntax.
- **“Performance loss after lesion proves phenomenal loss.”** It may reflect implementation damage, report failure, or regime collapse.
- **“An empty diagnostic set supports necessity because no counterexample exists.”** It leaves necessity unidentified.
- **“Synthetic ablation gives consciousness ground truth.”** It validates construct separability and engineering roles only.

Exit ticket

For N_2 , write the exact component profile on which the full model and the N_2 -ablated model disagree. Then name one forbidden dual-use measure.

Readings

Required: Stilwell, *Ablating N***, Sections 1-8 and 10, Appendices B and G.

Recommended: Doerig et al. (2021); Kleiner and Hoel (2021); Yaron et al. (2022); Cogitate Consortium (2025).

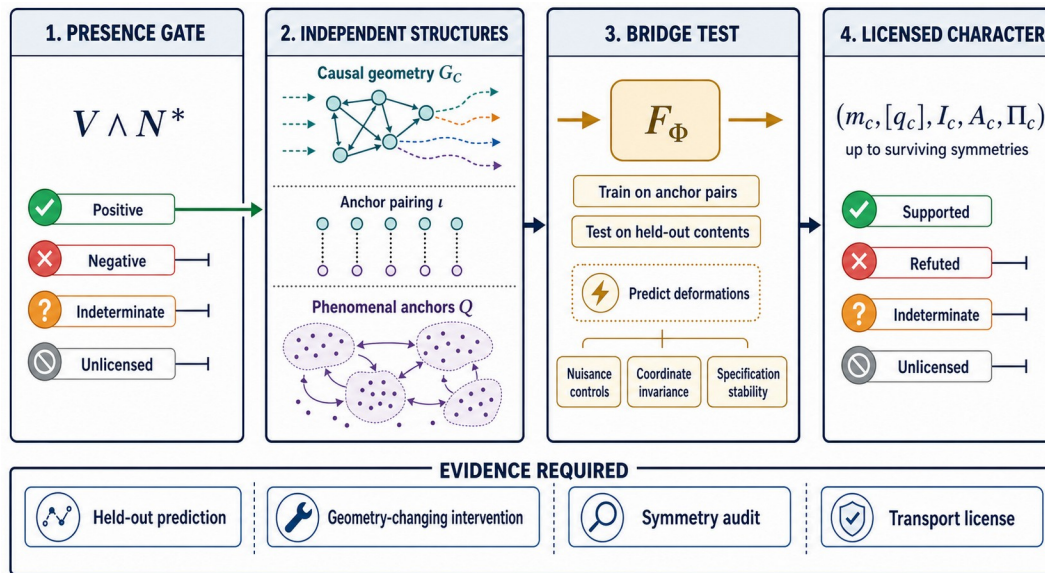
Session 9. From Phenomenal Presence to Phenomenal Character

Learning objectives

Students should be able to explain the two-layer architecture, distinguish causal-trajectory geometry from activation embeddings, describe the bridge-validation requirements, interpret character as a symmetry-qualified equivalence class, and separate bridge refutation from presence failure.

Instructor preparation

Assign *From Phenomenal Presence to Phenomenal Character*, Sections 1-5 and 7-9. Display the existing two-layer inference figure.



Two-layer inference architecture for presence and character

Timed plan

Time	Teaching move
0-15 min	Presence / character error-matrix warm-up
15-37 min	Lecture: two explananda and the presence gate
37-62 min	Lecture: causal-trajectory geometry and quality-space anchors
62-80 min	Bridge-fitting and deformation test
80-96 min	Symmetry, inversion, and transport
96-104 min	Modality applications
104-110 min	Exit ticket

Lecture notes

State the limitation that motivates the extension: $V \wedge N^*$ says whether experience is present, not why a present state feels red rather than green, painful rather than neutral, or continuous rather than stuttering.

The two-layer architecture is:

$$\text{presence gate} = V \wedge N^*,$$

$$\text{phenomenal character} = Q_\Phi(G_c, \Delta).$$

Presence is a gate, not another coordinate beside hue, pitch, intensity, or valence. Only a positive presence result licenses an ordinary phenomenal-character attribution. A rich geometry outside the gate remains scientifically meaningful representational structure, but the model does not call it phenomenal merely because it is structured.

Preserve both non-implications:

$$P(c_i) = P(c_j) = 1 \Rightarrow \not\Rightarrow \chi(c_i) = \chi(c_j),$$

and

$$G_{c_i} \cong G_{c_j} \Rightarrow \not\Rightarrow P(c_i) = P(c_j) = 1.$$

Define G_c carefully. It is not a single activation vector, a decoder embedding, or a convenient neural manifold. It is the content's neighborhood within a perturbation-derived geometry that contains:

- distances among content-specific response distributions;
- admissible causal transitions;
- trajectory timing, direction, and recurrence;
- occupancy under the declared regime; and
- intervention semantics and route sensitivity.

The geometry can be a metric space, pseudometric, directed graph, stratified manifold, or another relational structure. Do not force Euclidean structure when triangle, symmetry, or identity conditions fail.

Construct the phenomenal target independently. In report-capable anchor participants, use converging similarity judgments, discrimination thresholds, confusion matrices, generalization, adaptation, cross-context matching, and transition judgments. No single report defines the quality space. If anchor measures conflict beyond tolerance, the target geometry is indeterminate.

The bridge F_Φ maps frozen causal-geometry features to the independently estimated quality structure. Model family, dimensionality, nuisance controls, invariances, and thresholds belong to specification Φ and are selected within training data. A high in-sample correlation is insufficient.

A supported bridge should:

1. predict held-out phenomenal distances and neighborhoods better than declared nulls;
2. preserve transition structure and relevant asymmetries;
3. survive coordinate rotations, sensor relabeling, harmless rescaling, and reasonable preprocessing;
4. outperform nuisance models based on stimulus distance, semantics, arousal, or response preparation;
5. predict how geometry-changing interventions deform independently measured phenomenal relations; and
6. transport to new systems only under a separate bridge license.

The deformation test is the causal center. An intervention changes G_c in a preregistered direction. The frozen bridge predicts which phenomenal distances should contract, expand, or change transition order. The observed phenomenal deformation is collected independently. Baseline fit and intervention fit are reported separately so a good static correlation cannot hide a failed causal prediction.

Now introduce symmetry. If a transformation preserves every measured quality-space relation, the data cannot distinguish an absolute point q from its transformed counterpart. The warranted attribution is an equivalence class:

$$Q_{\Phi}(G_c, \Delta) = [F_{\Phi}(\psi_{\Phi}(G_c, \Delta))]_{Aut(Q)}.$$

This is the disciplined answer to inverted-spectrum concerns. A global red-green permutation remains underdetermined if it preserves every distance, transition, valence, memory relation, and causal role. Additional asymmetrical anchors may shrink the equivalence class, but the theory cannot assign an absolute label by assertion.

Summarize the structured character profile:

$$\chi_c = (m_c, [q_c], I_c, A_c, \Pi_c),$$

where m_c is modality stratum, $[q_c]$ relational location, I_c intensity, A_c valence field or vector, and Π_c temporal path profile. The tuple is not assumed exhaustive and does not require all modalities to occupy one commensurable Euclidean space.

The bridge outcomes are **supported**, **refuted**, **indeterminate**, and **unlicensed**. A refuted bridge in a positive presence case means the experience is present but this proposed geometry does not explain its differentiation. It does not mean there is no character. Conversely, a beautiful bridge fit cannot rescue an unlicensed presence gate.

Display plan

Use two separated streams:

$$\text{causal trajectories} \rightarrow G_c \parallel \text{phenomenal anchors} \rightarrow Q$$

followed by a bridge trained on anchors and tested on held-out contents and interventions. Draw no arrow from quality fit back to presence classification.

Slide sequence

1. Presence is not character
2. The two non-implications
3. Presence-gated domain
4. What G_c contains
5. What G_c is not
6. Independent quality-space anchors
7. The bridge F_{Φ}
8. Held-out prediction
9. Geometry-changing intervention test
10. Symmetry-qualified character
11. Modality, location, intensity, valence, path
12. Four bridge outcomes

Active-learning exercise: Refuting a bridge without denying experience

Students receive a hypothetical color experiment in which $V \wedge N^*$ remains positive, V4-like causal geometry changes under adaptation, but reported and low-demand phenomenal similarity relations do not change. They decide which part of the program is challenged, list nuisance explanations, and specify what additional result would make the bridge refuted rather than indeterminate.

Discussion prompts

1. Does a lawlike geometry-to-quality mapping explain character or merely predict it?

2. Is identifying character only up to automorphism an adequate theory of qualia?
3. Could valence or meaning require nongeometric structure?

Common misconceptions

- **“Any neural manifold is a quality space.”** The predictor and phenomenal target must be independently constructed and causally bridged.
- **“A failed bridge means the state was unconscious.”** Character mapping and presence classification are modular.
- **“Similarity reports make the theory report-dependent.”** Reports are one anchor family in calibration cases, not the presence criterion.
- **“A relational theory rules out inversion.”** The framework explicitly preserves unbroken symmetries.

Exit ticket

State one reason decoding accuracy is weaker than bridge fit and one result that would refute Q_ϕ while leaving N^* intact.

Readings

Required: Stilwell, *From Phenomenal Presence to Phenomenal Character*, Sections 1-5 and 7-9.

Recommended: Fleming and Shea (2024); Kriegeskorte and Kievit (2013); Brouwer and Heeger (2009); Clark (1993).

Session 10. Reading Consciousness from the Schematic

Learning objectives

Students should be able to distinguish correlational, black-box, and constitutive readings of the theory; reconstruct the role vector; apply the four schematic-legibility conditions; and evaluate the analogy's force and limits.

Instructor preparation

Assign *Reading Consciousness from the Schematic* in full. Bring a simple electrical schematic or another expert-readable mechanistic diagram whose operating state changes under a localized fault.

Timed plan

	Time	Teaching move
	0-18 min	What does an engineer read from a schematic?
	18-40 min	Lecture: black-box versus structurally legible necessity
	40-62 min	Lecture: bearer, operation, unity, system-level presence, temporal presence
	62-78 min	Legibility test: D, C, F, E
	78-96 min	Objections: circularity, zombies, observation, and functionalism
	96-104 min	Teachability as an empirical standard
	104-110 min	Exit ticket

Lecture notes

The schematic paper begins after a conditional concession: suppose the biconditional survives empirical testing. Would it remain an opaque law that simply pairs a dynamical bundle with consciousness, or could a trained observer understand the organization as the phenomenal state itself?

Contrast:

black-box necessity : $X \Rightarrow C_0$

with

structurally legible necessity : $X \Rightarrow \rho \Rightarrow C_0$.

ρ is an interpretable role profile. For CØ / N* it contains:

$\rho_{C_0} = \langle \text{bearer, realized operation, unity, system-level presence, temporal presence} \rangle$.

Boundary qualification supplies the bearer. V supplies realized operation. N_1 supplies unity. N_2 supplies system-level causal presence. N_3 supplies bounded temporal presence. Under the constitutive hypothesis, this role-complete organization is not merely a cause or correlate of C_0 ; it is proposed as the physical form of C_0 .

Use the electrical analogy carefully. An engineer sees more than ink because symbols have learned component semantics, component laws compose, inputs and operating conditions are declared, and counterfactual faults have localized consequences. The diagram alone neither powers the circuit nor validates the component laws. Likewise, an N^* diagram neither experiences nor proves the identity. The actual system must realize the dynamics, and independent evidence must support the constitutive hypothesis.

Define schematic legibility:

$$\text{Legible}(M, o / K) \Leftrightarrow D \wedge C \wedge F \wedge E.$$

- **Decomposable semantics (D):** elements have stable role meanings not defined by repeating the target outcome.
- **Compositional readout (C):** a trained reader can derive an intermediate organization profile and apply an explicit rule.
- **Counterfactual fault localization (F):** licensed changes support differentiated predictions about which aspect fails or changes.
- **Evidential independence (E):** the readout procedure is not counted as evidence that its own constitutive rule is true.

The fault map is central. Boundary failure makes the inference unlicensed. Viability failure removes the operating regime. Selective N_1 failure removes unified content for that candidate. Selective N_2 failure leaves content causally isolated. Selective N_3 failure leaves it temporally evanescent. A change in G_c under an intact presence gate alters predicted character rather than presence. These are differentiated failures, not one generic low score.

Teach the three strength levels:

Position	Claim	Explanatory status
Correlational marker	N^* covaries with consciousness evidence	Useful prediction; organization may be incidental
Black-box law	$V \wedge N^*$ reliably guarantees C_0	Strong law; the transition remains opaque
Constitutive schematic	C_0 is the role-complete organization	The trained reader reconstructs presence and its failure map

The strong reading is an a posteriori identity claim, not a dictionary definition. Its opponents can understand it perfectly and deny that the identity is true. Structural intelligibility is not confirmation.

Address the zombie objection. A physical duplicate without experience would defeat sufficiency if independently established. Merely imagining the duplicate or noting the absence of first-person access does not provide that evidence. The substantive dispute is whether the complete organization is constitutive.

Address external observation. Distinguish observing that another system instantiates C_0 , inferring features of its token character q_S , and undergoing q_S from that system's first-person position. A scientific observer need not become the other subject. If the identity is true and supported, observing the realized organization is observing consciousness under a public causal-dynamical description, even though first-person acquaintance remains indexed to the bearer.

Address functionalism. The proposal is a physicalist information-dynamical functionalism, not unrestricted input-output equivalence. Static diagrams, behavioral lookup tables, and merely isomorphic software descriptions do not qualify. Effective causal realization, correct grain, boundary, time, and viability do real work.

End with teachability as a standard. Independent investigators should learn the readout rules, agree on held-out cases under the same frozen specification, predict localized failures, and transport the role profile across materially different implementations when causal organization is preserved. If only the author can "see" the identity, it is not schematic legibility.

Display plan

Draw:

$$S_{N^*} \rightarrow \rho_{C_0} \rightarrow C_0$$

Label the first arrow **mechanistic interpretation** and the second **exposed identity hypothesis**. Underline that only independent evidence bears on the truth of the second arrow.

Slide sequence

1. The remaining explanatory lacuna
2. What an expert reads in a schematic
3. The N^* schematic
4. The role vector
5. Decomposable semantics
6. Compositional readout
7. Counterfactual fault localization
8. Evidential independence
9. Marker, law, or constitution?
10. Observation versus undergoing
11. Limits of the analogy
12. Teachability and transport

Active-learning exercise: Legible or merely labeled?

Give groups four models: a one-number classifier labeled “consciousness”; a component diagram with no counterfactual predictions; a detailed N^* profile whose validation uses the same scores as labels; and a frozen role model with independent anchors and fault predictions. Students apply D , C , F , and E and explain which condition fails.

Discussion prompts

1. Does structural legibility narrow the explanatory gap or simply restate functionalism more clearly?
2. If two trained readers agree on the readout but disagree about the identity, what has the schematic achieved?
3. Can consciousness be publicly observed without first-person acquaintance?

Common misconceptions

- **“The schematic proves consciousness visually.”** A representation is not a realization, and readability is not confirmation.
- **“Role language smuggles phenomenality into the premises.”** The roles are operationalized without a consciousness verdict; the identity supplies the phenomenal claim.
- **“First-person privacy makes all third-person evidence defective.”** It limits the mode of access, not necessarily the possibility of warranted attribution.
- **“The circuit analogy claims equal scientific maturity.”** It describes a target epistemic form, not current evidential parity.

Exit ticket

State the difference between a black-box law and a constitutive schematic, and name one way the latter could fail empirically even if it remains intelligible.

Readings

Required: Stilwell, *Reading Consciousness from the Schematic*.

Recommended: Machamer, Darden, and Craver (2000); Craver (2007); Levine (1983); Chalmers (1995).

Session 11. Biological and Clinical Applications

Learning objectives

Students should be able to apply the full inference order to variable human states, distinguish target-specific from whole-system conclusions, and identify the major validity risks in report-limited clinical and experimental cases.

Instructor preparation

Assign the worked-case sections of the core, boundary, availability, indeterminacy, and ablation papers. Divide students into clinical, sleep/anesthesia, perceptual, and split-system teams.

Timed plan

	Time	Teaching move
	0-18 min	Instructor demonstration: one complete classification tuple
	18-35 min	Waking, REM, NREM, and anesthesia
	35-54 min	Locked-in, unresponsive wakefulness, and minimally conscious states
	54-70 min	Blindsight and binocular rivalry
	70-90 min	Split-brain boundary analysis
	90-103 min	Team case conference
	103-110 min	Exit ticket

Lecture notes

Require the same order for every case:

1. specify the target C ;
2. declare the candidate system, grain, regime, and interval;
3. audit boundary status;
4. assess V independently;
5. estimate N_1 , N_2 , and N_3 for the same target;
6. apply conflict and uncertainty rules;
7. state the model output; and
8. separate that output from clinical or ethical action.

Use ordinary waking perception only as a calibration case. It should not define consciousness by report. For a dominant percept in a healthy adult, a stable boundary, viable regime, integration, availability, and recurrence are expected. The scientific advantage is abundant anchor evidence; the danger is overfitting every operationalization to report-capable adults and assuming transport.

REM dreaming. Concurrent motor report is absent and external interaction is reduced. The candidate can still satisfy an internally defined N_2 if dream contents causally modulate diverse internal recipients. Retrospective reports are temporally displaced and memory-dependent. A positive result requires dream-state criteria for availability and recurrence, not waking executive criteria imposed by default.

Deep NREM sleep. The model allows sparse, episodic, content-specific, or indeterminate presence. A single negative awakening report cannot establish dreamlessness because memory and timing can fail. Repeated state sampling,

perturbational measures, and carefully timed awakening anchors are needed. Different intervals in one night may receive different outputs.

General anesthesia. Agents should not be pooled as one mechanism. Some primarily alter thalamocortical organization, some recurrent connectivity, and some neuromodulatory regimes. The model predicts loss of C_0 when the bundle collapses, but a valid test must distinguish viability failure, component failure, and unlicensed measurement. Recovery and dreaming-like anesthesia experiences are important near-miss cases.

Locked-in syndrome. Motor report failure is not a consciousness failure. A stable boundary and viable N^* bundle can support a positive prediction when speech and movement are absent. The case is a direct reason for report-independent N_2 .

Unresponsive wakefulness syndrome and cognitive motor dissociation. Behavioral non-response has many possible causes. Task-based fMRI or EEG can reveal command following in some behaviorally unresponsive patients. A positive command-following result supports at least some awareness-relevant capacity but does not automatically reveal continuity or richness. A negative task result is licensed only when hearing, comprehension, arousal, attention, memory, and motor-independent response generation had a fair opportunity to operate. Repeated assessment matters because states fluctuate.

Minimally conscious state. The framework expects time- and content-indexed variability. A single negative window should not be generalized. Report the distribution of positive, negative, indeterminate, and unlicensed windows along with causes.

Blindsight. Accurate forced choice in a blind field does not by itself establish visual C_0 . The theory predicts no conscious blind-field content if the relevant integrated, available, recurrent bundle fails, while allowing unconscious information to guide behavior. The important comparison is not performance versus no performance but the causal component profile for the content.

Binocular rivalry. Physical stimulation can remain stable while perceptual content alternates. Involuntary markers can time the dominant content without requiring trial-by-trial report. The model predicts that content-specific N^* tracks the dominant percept. Eye movement is an anchor or timing aid, not a constituent. Physical-alternation controls and report/no-report contrasts separate perceptual, introspective, and motor activity.

Split brain. The case first tests boundaries, not the count of reports. Evaluate left, right, and residual joint candidates. If each side becomes autonomous, role-complete, stable, and pairwise separable, multiple candidate bearers are supported. If shared subcortical recurrence preserves a joint candidate or results vary by task and horizon, report a larger, layered, or boundary-indeterminate result. Do not compare pre- and post-disconnection component scores as one persistent subject unless boundary identity is preserved.

Use the following summary matrix as hypotheses, not diagnoses:

Case	Initial component expectation	Proper caution
Waking dominant percept	$V +, N^* +$	Baseline anchors do not define the theory by report
REM dream	$V +$ under dream regime; N_1, N_3 plausible; N_2 internal	Retrospective report and memory are displaced
Deep NREM	fluctuating, sparse, or unresolved	One negative awakening report is weak
Adequate anesthesia	multiple component collapse likely	Agent, dose, timing, and covert experience vary
Locked-in syndrome	$V +$ and N^* may remain	Motor silence is not N_2 failure
Unresponsive wakefulness	variable and often unresolved	Negative license and repeated assessment are essential

Case	Initial component expectation	Proper caution
Blindsight	content-specific bundle may be incomplete	Forced choice is access-like evidence, not decisive presence
Binocular rivalry	bundle should track dominant content	Involuntary markers time content but do not constitute it
Split brain	one, multiple, layered, or unresolved candidates	Anatomy alone does not count subjects

Display plan

Maintain one fixed case worksheet with rows for target, boundary, V , N_1 , N_2 , N_3 , conflicts, output, cause profile, and action note. Complete it in public for one case before team work.

Slide sequence

1. The fixed inference order
2. Waking as calibration, not definition
3. REM dreaming
4. Deep NREM
5. Anesthesia by mechanism and window
6. Locked-in syndrome
7. Disorders of consciousness
8. Blindsight
9. Binocular rivalry
10. Split-brain boundaries
11. Case matrix

Active-learning exercise: Multidisciplinary case conference

Each team receives a case with incomplete or conflicting results. One student acts as boundary specialist, one as component-measure specialist, one as epistemic auditor, and one as ethics/clinical liaison. The team produces a six-place assessment tuple, an outcome, a cause profile, and a distinct practical recommendation.

Discussion prompts

1. Which clinical false error is more dangerous: false positive or false negative?
2. Should retrospective dream reports be allowed to calibrate online no-report measures?
3. Can a split-system result change through task and time without implying unstable metaphysics?

Common misconceptions

- **“Wakefulness is phenomenal presence.”** Arousal is a regime variable, not the target.
- **“Command following measures the full richness of experience.”** It supports a narrower capacity and depends on auxiliary demands.
- **“No-report paradigms are confound-free.”** Involuntary markers have their own false positives and target assumptions.
- **“Split brain means two subjects by anatomy.”** Multiplicity requires causal autonomy, role completeness, and separability.

Exit ticket

Choose one clinical or experimental case and identify the single most likely route to a false negative. State the control required for a licensed negative inference.

Readings

Required: Worked-case sections from all six empirical companion papers.

Recommended: Bodien et al. (2024); Wannez et al. (2017); Frassle et al. (2014); Tsuchiya et al. (2015).

Session 12. Nonhuman Animals, Organoids, and Artificial Systems

Learning objectives

Students should be able to explain constrained substrate independence, conduct a transport audit, avoid both biological chauvinism and syntactic overattribution, and formulate a precautionary action without changing the scientific status.

Instructor preparation

Assign the nonhuman, organoid, and AI applications in the core, boundary, availability, indeterminacy, and ablation papers. Ask students to choose one target system and diagram its actual running architecture or organismic causal organization.

Timed plan

Time	Teaching move
0-20 min	Lecture: substrate independence with teeth
20-38 min	Animals and developmental transport
38-56 min	Organoid-computer hybrids
56-75 min	Artificial and distributed architectures
75-93 min	Transport-audit workshop
93-104 min	Welfare and precaution under indeterminacy
104-110 min	Exit ticket

Lecture notes

The theory is substrate-independent in form but not permissive in application. A cross-substrate mapping must preserve the relevant causal-dynamical roles, not merely a diagram or input-output function. At minimum it must preserve directionality, effective coupling, temporal grain, recurrent organization, functional recipient structure, dynamic parameter ranges, and substrate-appropriate viability within preregistered tolerances.

This position avoids two symmetrical errors. **Neural chauvinism** denies consciousness to any non-neural realization without asking whether the causal role is preserved. **Empty functionalism** grants consciousness to any behavioral or abstract isomorph. CØ / N* requires actual realized organization under a validated mapping.

Teach the three transport bridges:

1. **Construct preservation:** does the measure still detect the same functional or dynamical property?
2. **Causal-role preservation:** does that property play the same theoretically relevant role in the new system?
3. **Calibration preservation:** do thresholds, error rates, and uncertainty rules remain defensible?

Failure of any bridge produces transport indeterminacy or an unlicensed application. Running the same code or formula on a new substrate does not carry the old interpretation.

Nonhuman animals and infants. Recipient classes, viability indicators, intervention burdens, and anchors must be species- and stage-appropriate. Failure to learn a human-designed task can reflect ecology, sensory differences, motivation, or training demand. Use homologous causal roles, flexible behavior with known sensory dependencies, motivational tradeoffs, learning and generalization, perturbational predictions, and converging physiological evidence. A mammalian neural marker can be informative in another species only with an explicit bridge.

Organoids. Neural activity, oscillation, complexity, and perturbational response can be measured, but their interpretation was developed in organisms with differentiated anatomy, sensory history, vascular and regulatory support, and behavior.

The first question is the candidate boundary: tissue alone, tissue-electrode loop, tissue plus controller, or the full apparatus. If a digital controller supplies indispensable recurrence or memory and generic replacement fails, a hybrid boundary may be appropriate. If the controller imposes the trajectory, the assembly may fail autonomy. None of these boundary results is itself a consciousness verdict.

Artificial systems. Inspect the running architecture, not the product label. A base model, prompt, context window, retrieval service, persistent memory, scheduler, tools, evaluators, other models, and human operators can form different candidates at different horizons. A stateless model call may have complex internal transformation but little recurrent continuity. An agent loop can add persistence and availability while remaining externally scripted. A multi-agent service may contain several autonomous candidates, a larger workspace-like candidate, or no role-complete candidate.

The proper tests include state resets, memory substitution, scheduler ablation, message delay, route interruption, service removal, human-in-the-loop removal, latent or token interventions, and matched-controller replacement. Logging services, mirrored replicas, idle subscribers, and fluent language do not automatically add integration or functional diversity.

Use artificial systems first as construct laboratories. They allow precise criterion, mechanism, and measurement ablations and can reveal whether proposed N_i measures behave as intended. This validates the measurement architecture, not consciousness. Phenomenal interpretation still requires independent anchors and transport discipline.

Ethical decisions remain separate. An indeterminate animal, organoid, or AI assessment can justify documentation, monitoring, harm minimization, scaling limits, or independent review when potential harm is serious and irreversible. The practical action reflects evidence plus a declared loss structure. It must not be reported as scientific proof of consciousness.

Display plan

Draw a transport bridge with three load-bearing supports: **construct**, **causal role**, and **calibration**. Remove one support and label the result **unlicensed transport**, not negative.

Slide sequence

1. Substrate independence with teeth
2. Neural chauvinism and empty functionalism
3. Three transport bridges
4. Species-appropriate recipient classes
5. Developmental and ecological demands
6. Organoid boundary alternatives
7. Hybrid constitution versus imposed control
8. Actual AI running architectures
9. Service and state perturbations
10. Construct validation versus consciousness evidence
11. Precaution without relabeling

Active-learning exercise: Transport audit

Teams choose one measure validated in awake adults and propose its use in an octopus, infant, organoid, or artificial agent. They complete a transport ledger with construct, causal-role, and calibration evidence; name the most likely failure; assign the present scientific status; and propose the smallest next study that would improve the license.

Discussion prompts

1. Can artificial systems ever satisfy a meaningful viability condition without biological self-maintenance?

2. When does an external memory become a constituent rather than an interface?
3. Should ethical thresholds for precaution be lower than scientific rule-in thresholds?

Common misconceptions

- **“Substrate-independent means software equivalence is enough.”** The mapping must preserve realized causal dynamics and operating ranges.
- **“Biological markers absent in AI prove no consciousness.”** An out-of-domain biological test may be unlicensed rather than negative.
- **“A brain-like number in an organoid has the same meaning.”** Measurement and causal-role transport must be established.
- **“Precaution is evidence.”** It is a policy response to evidence and asymmetric risk.

Exit ticket

Name one feature that must be preserved for cross-substrate mapping and one reason that applying a human-calibrated test to a new substrate could be unlicensed rather than negative.

Readings

Required: Nonhuman, organoid, and AI application sections across the paper corpus.

Recommended: Butlin et al. (2023); Birch et al. (2020); Kagan et al. (2022); Smirnova et al. (2023).

Session 13. Differentiation, Validation, and the Strongest Objections

Learning objectives

Students should be able to state what is genuinely distinctive about CØ / N*, organize evidence into validation layers, compare the model fairly with neighboring theories, and identify observations that would require revision rather than reinterpretation.

Instructor preparation

Assign the theory-comparison and objections sections of all seven papers. Ask each student to prepare the strongest version of one objection, not a rhetorical caricature.

Timed plan

Time	Teaching move
0-20 min	Lecture: the distinctiveness argument
20-42 min	Lecture: validation ladder and evidential dependencies
42-62 min	Comparative prediction matrix
62-86 min	Steelman objection carousel
86-101 min	Revision rules and research-program health
101-110 min	Exit ticket

Lecture notes

Begin with the obvious challenge: the components are not individually novel. The distinctive proposal lies in their **division of labor and joint empirical discipline**. The theory says:

1. the target is minimal phenomenal presence rather than report or self-reflection;
2. one frozen candidate must realize all roles in one window;
3. integration, availability, and recurrence are jointly required and individually insufficient;
4. viability is independently assessed and cannot be retrofitted;
5. availability is causal, functionally diverse, and report-independent;
6. unresolved credible conflict blocks a positive verdict;
7. the biconditional is vulnerable in both directions;
8. every conjunct competes with leave-one-out and grouped rivals on diagnostic cases;
9. character is a separately validated second layer; and
10. the intended identity has a compositional fault map rather than an unexplained label.

This is more than “IIT + GNWT + RPT” because source theories do not jointly force these restrictions. IIT does not ordinarily require separate report-independent broadcast and bounded recurrence. GNWT does not ordinarily require synergy plus a separate character bridge. RPT does not ordinarily require functionally diverse availability. None by itself supplies the boundary-first autonomy procedure, four-outcome precedence policy, diagnostic-set audit, or schematic-legibility standard.

Present a validation ladder. Higher rungs presuppose lower ones:

1. **Conceptual clarity:** targets, roles, scope, and exclusions are explicit.

2. **Construct separability:** measures can distinguish integration, availability, recurrence, and viability in synthetic and benchmark systems.
3. **Measurement validity:** surrogates track their intended roles rather than activity, report, arousal, or one latent factor.
4. **Boundary validity:** candidate systems are independently individuated and stable.
5. **Bundle discrimination:** $V \wedge N^*$ predicts anchor patterns across states and contents.
6. **Component necessity:** full models beat proper submodels on adequate diagnostic coverage.
7. **Causal intervention:** selective perturbation and rescue alter independent anchors as preregistered.
8. **Transport:** roles and thresholds generalize without favorable retuning.
9. **Character bridge:** causal geometry predicts held-out phenomenal relations and deformations.
10. **Schematic competence:** independent investigators can read, fault-localize, and transport the organization.

Stress that success at one rung does not automatically establish the next. A synthetic network can validate construct separability without consciousness. A state-level classifier can support bundle discrimination without component necessity. A character bridge can fail while presence remains supported.

Use a discriminating comparison matrix:

Case	C_0 / N^*	IIT-style integration sufficiency	GNWT-style access emphasis	RPT-style recurrence sufficiency
High integration, local and unstable	No C_0 prediction	Some versions may favor consciousness	No global access	No stable recurrence
Broad report-free reach, low integration	No C_0 prediction	Integration requirement fails	Availability may favor consciousness	Recurrence status decides
Stable local loop, low reach	No C_0 prediction	Depends on integration	No broad access	Some versions may favor consciousness
Motor report loss with intact bundle	C_0 predicted	Depends on causal structure	Report-focused implementations risk false negative	May preserve consciousness
Rich unconscious geometry	No ordinary character attribution without gate	IIT may treat causal structure as phenomenal	Content access may be absent	Recurrence may or may not occur
Full bundle with strong negative anchors	Sufficiency threat	Theory-specific response	Theory-specific response	Theory-specific response

The matrix must be tailored to precise versions of rival theories; it is a teaching device, not a substitute for reading them.

Steelman the major objections.

Hybrid without novelty. Reply with the exposed conjunction and methodological package, then admit that distinctiveness must yield predictions no source theory already forces.

Access smuggling. Reply that N_2 is internal, counterfactual, content-specific causal availability, and must pass without report or executive classes. Admit that the objection succeeds if every credible implementation requires those classes.

Viability escape hatch. Reply with independent preregistration and versioned rules. Admit that repeated bespoke revisions damage transport credibility.

Threshold arbitrariness. Reply that explicit, stress-tested thresholds are scientifically preferable to hidden ones. Admit that brittle classifications should remain fragile or indeterminate.

Correlation rather than constitution. Reply that interventions, route tests, ablations, and rescue matter more than passive decoding. Admit that even causal relevance does not by itself prove metaphysical identity.

Hard problem and explanatory gap. Reply that the program offers a mid-level constitutive identity and structural legibility, not a derivation of why any physical organization is accompanied by experience. The metaphysical objection remains live.

Embodiment and affect underplayed. Reply that boundary, V , content dynamics, and the character profile can include bodily and affective organization. Admit that future evidence may require an organismic or affective conjunct.

Conjuncts are causally inseparable. Reply that the audit includes a causally coupled outcome. If inseparability is robust, the theory should revise its factorization rather than advertise separately identified necessity.

No consciousness oracle. Reply that held-out convergent anchors permit comparative empirical tests without pretending to direct observation. Admit that the results remain anchor-relative and cannot close the other-minds gap.

End with revision rules. A healthy research program specifies what changes after failure: remove a redundant conjunct; combine coupled components; narrow a transport claim; revise an invalid measure; replace the sharp biconditional with a probabilistic rule if threshold instability is systematic; add a conjunct if repeated diagnostic cases demand it; or reject the bundle if licensed counterexamples persist.

Display plan

Draw the validation ladder. On the side write: **No upward inference without the lower license.**

Slide sequence

1. What is genuinely distinctive?
2. More than a hybrid list
3. The ten-rung validation ladder
4. Comparative near-miss profiles
5. Hybrid-without-novelty objection
6. Access and report objection
7. Viability and threshold objections
8. Correlation / constitution objection
9. Hard problem and embodiment objections
10. Coupling and oracle objections
11. Revision rules
12. What would make the program progressive?

Active-learning exercise: Objection carousel

Pairs rotate through four stations: conceptual target, empirical measurement, theory comparison, and metaphysics. At each station they improve an objection, state the best reply available in the corpus, and identify the residual pressure the reply does not remove. The goal is a balanced argument map, not a winner.

Discussion prompts

1. Which part of the program is currently most vulnerable: target, components, measurement, identity, or transport?
2. What unique successful prediction would most strongly distinguish the conjunction?
3. When does a flexible research program become unfalsifiable?

Common misconceptions

- **“A theory is novel only if every component is new.”** New constraint architecture and predictions can be theoretically substantive.
- **“Falsifiability requires a consciousness oracle.”** Comparative risky predictions can be tested against independent anchor structures, though not with metaphysical certainty.
- **“Every objection has been answered.”** Several replies explicitly preserve residual philosophical and empirical pressure.
- **“A component failure destroys every paper.”** The program is modular; some failures motivate revision rather than total collapse.

Exit ticket

State the strongest differentiating claim of $C\emptyset / N^*$ and one empirical result that would force a nontrivial revision of that claim.

Readings

Required: Theory-comparison and objection sections across all seven papers.

Recommended: Seth and Bayne (2022); Mudrik et al. (2025); Doerig et al. (2021); Cogitate Consortium (2025).

Session 14. Capstone Adversarial Research Design

Learning objectives

Students should be able to integrate the complete research workflow, design a study with genuine rival predictions, define positive, negative, indeterminate, and unlicensed outcomes in advance, and defend a revision rule.

Instructor preparation

Distribute the Capstone Preregistration Template in Appendix C two weeks in advance. Assign teams with deliberately mixed theoretical preferences. Require a one-page rival-model memorandum before class.

Timed plan

Time	Teaching move
0-12 min	Recap: the complete inference architecture
12-28 min	Instructor model of a good adversarial question
28-70 min	Team preregistration clinic
70-94 min	Oral defenses before a skeptical panel
94-104 min	Cross-team revision commitments
104-110 min	Final synthesis

Lecture notes

The complete workflow is:

target $\rightarrow$ candidate boundary $\rightarrow V \rightarrow (N_1, N_2, N_3) \rightarrow$ conflict and validity gates $\rightarrow$ presence output $\rightarrow$ component aud

A capstone project should not attempt every element. It should select one tractable point where a result would alter the model. Good projects include:

- an N_2 diagnostic case that preserves N_1 and N_3 ;
- a feedforward long-timescale adversary for N_3 ;
- a boundary intervention in a hybrid biological-digital system;
- a grouped ablation that tests whether N_1 and N_3 are jointly redundant;
- a transport audit of one surrogate across species or substrate;
- a geometry-changing intervention that differentiates causal-trajectory and stimulus-only models; or
- a reproducibility study of schematic readout across independent analysts.

Require every team to state:

1. the exact target and unit of analysis;
2. the full and rival claims;
3. the cases on which the claims disagree;
4. the boundary, grain, regime, and interval;
5. the discovery, calibration, and final-test split;
6. component estimators and uncertainty bounds;

7. independent anchors and their validation genealogy;
8. intervention selectivity and retained-component equivalence margins;
9. rule-in, rule-out, conflict, indeterminate, and license-failure conditions;
10. sensitivity and transport analyses;
11. the result each team member expects; and
12. the revision that follows from every declared outcome.

A good adversarial question has three properties. First, rival predictions differ before the result. Second, both sides consider the test fair. Third, the outcome cannot be absorbed by moving the boundary, target, threshold, viability criterion, or anchor set after inspection.

Model one example: a recurrent visual content is stabilized locally and remains integrated, but a reversible route intervention prevents modulation of diverse recipients while preserving report-independent content anchors. The full model predicts loss or degradation of presence-like anchors when N_2 fails. The N_2 -ablated rival predicts that anchors remain. The study must verify that the source content, N_1 , N_3 , V , and boundary are stable; report and motor pathways cannot be the decisive anchor; and the route intervention cannot create a new regime.

The skeptical panel should ask:

- Could the boundary change?
- Is the target narrower than the conclusion?
- Does the intervention selectively move the intended component?
- Is any measure used twice?
- Is there adequate diagnostic coverage?
- Could common input, arousal, task demand, report, or sensor leakage explain the result?
- What output occurs if thresholds straddle?
- What observation would make the team abandon or revise its favored interpretation?

Close by returning to the course stance. A mature understanding of a new theory is not the ability to repeat its slogans. It is the ability to expose its commitments, preserve its inferential limits, and design the fair test its proponents and critics should both want.

Display plan

Keep a visible checklist with the twelve required fields. Place a red mark beside any element that a team attempts to choose after seeing the critical result.

Slide sequence

1. Complete inference architecture
2. Choose one tractable pressure point
3. Full and rival predictions
4. Diagnostic cases
5. Freeze the specification
6. Split the evidence
7. Protect anchor independence
8. Define four outcomes
9. Sensitivity and transport

10. Revision rules
11. Skeptical-panel questions

Active-learning exercise: Adversarial preregistration clinic

Each team presents a five-minute design. Another team acts as the ablated-rival group and identifies one route to circularity, one validity failure, and one result that the original team has not classified. The presenting team revises its protocol in writing before the final oral defense.

Discussion prompts

1. Which model commitment is most experimentally tractable now?
2. Which commitment is most scientifically important but least tractable?
3. What would count as evidence that the whole research program, rather than one measure, is failing?

Common misconceptions

- **“A capstone must test consciousness directly.”** A rigorous construct, boundary, transport, or component audit can make a decisive contribution.
- **“More measures guarantee independence.”** Measures can share sensors, preprocessing, task demands, and theoretical assumptions.
- **“An indeterminate result means the study failed.”** A preregistered cause-specific indeterminate result can identify the next discriminating experiment.
- **“Revision rules signal weakness.”** They distinguish a scientific program from a protected doctrine.

Final exit ticket

Write the one result you think would most increase confidence in $C\phi$ / N^* and the one result you think would most decrease it. For each, state the inferential license required before it counts.

Readings

Required: Review all protocols and preregistration appendices in the seven-paper corpus.

Recommended: Melloni et al. (2023); Cogitate Consortium (2025); Yaron et al. (2022).

Part III. Instructor Argument Bank

This section gives compact, classroom-ready arguments for differentiating and evaluating the theory. Each argument includes the claim, its rationale, the evidence needed, and the principal residual pressure. These are arguments for a research program, not declarations that the empirical burden has already been met.

Argument 1. Explanandum discipline

Claim. CØ / N* improves theory comparison by fixing minimal phenomenal presence as its initial target and refusing to identify it with report, attention, intelligence, wakefulness, self-consciousness, or detailed character.

Rationale. Many apparent disputes compare theories of different explananda. A target-first framework prevents report-related or metacognitive mechanisms from being credited as necessary for minimal presence without a separate argument.

Validation route. Show that component measures track presence-relevant contrasts after report, executive demand, and overt behavior are controlled, and that richer capacities can dissociate from the minimal bundle.

Residual pressure. The target may be too thin to satisfy those who regard ownership, affect, or qualitative structure as constitutive of any experience.

Argument 2. Conjunctive differentiation

Claim. Integration, availability, and recurrence solve different false-positive problems, so a conjunction can outperform single-factor theories.

Rationale. Integration blocks heaps, availability blocks isolated contents, and recurrence blocks transients. Each condition alone permits a characteristic near miss.

Validation route. Enrich for single-failure diagnostic cases and compare the full model with every leave-one-out and scientifically motivated grouped rival on independent anchors.

Residual pressure. The conditions may be manifestations of one deeper process or too causally coupled to warrant separate necessity claims.

Argument 3. Boundary-before-measurement non-circularity

Claim. Any theory using system-level quantities requires an independently motivated candidate system before those quantities can support a consciousness verdict.

Rationale. Integration, availability, recurrence, complexity, and causal power all change with the selected node set, grain, and horizon. Choosing the boundary that maximizes the final score lets evidence select its own bearer.

Validation route. Demonstrate that autonomy-first boundaries recover stable causal units, predict held-out perturbations, and outperform anatomical, product-label, random, and score-maximizing baselines.

Residual pressure. The autonomy measure, proposal family, and inclusion-minimality rule may miss a genuinely larger or cross-scale subject.

Argument 4. Report-independent availability

Claim. System-wide availability can be operationalized as a causal disposition without making report or executive access constitutive.

Rationale. Content-specific interventions can reveal timely modulation of functionally diverse recipients even when no output is produced. Diffusion, common input, arousal, and copied messages can be excluded by selectivity, route, and diversity controls.

Validation route. Show that N_2 remains stable in no-report, motor-dissociation, posterior-dominant, and nonexecutive conditions while unconscious diffusion and message duplication fail.

Residual pressure. If all reliable recipient effects require memory, decision, executive control, or report, the proposed separation from access consciousness fails.

Argument 5. Epistemic discipline increases rather than weakens falsifiability

Claim. Positive, negative, indeterminate, and unlicensed outputs make empirical claims more accountable by preventing missing or invalid evidence from being converted into favorable verdicts.

Rationale. A sharp theory can coexist with a limited test. Validity and quality gates, a preregistered conflict rule, and protected gray zone prevent cherry-picking and false negative inflation.

Validation route. Report coverage, selective risk, indeterminate rate, unlicensed rate, cause profiles, and calibration over repeated assessments. Show that abstention improves conditional error without hiding systematic failures.

Residual pressure. Indeterminacy can become a rescue category unless thresholds, applicability, conflict rules, and maximum acceptable unresolved rates are set prospectively.

Argument 6. The necessity audit prevents syntactic self-confirmation

Claim. A component earns its place only when the full and ablated models differ on valid diagnostic cases and the full model predicts independent held-out anchors better.

Rationale. A term is logically necessary inside a written conjunction but may be empirically redundant. Diagonal datasets cannot distinguish components. Labels generated by N^* cannot validate N^* .

Validation route. Establish diagnostic coverage, preserve anchor independence, verify selective mechanism changes, use matched model flexibility, test grouped ablations, and transport without favorable retuning.

Residual pressure. Anchor evidence remains theory-laden and diagnostic conditioning can create selection bias.

Argument 7. Modular extension to phenomenal character

Claim. Separating presence from character allows the theory to explain phenomenal differentiation without letting representational geometry manufacture consciousness.

Rationale. A presence gate and an independently tested geometry-to-quality bridge answer different questions and can fail independently.

Validation route. Fit the bridge on independent anchors, predict held-out relations and intervention-induced deformations, beat nuisance models, and report unbroken automorphisms and transport limits.

Residual pressure. Relational geometry may not exhaust intrinsic, semantic, affective, bodily, or historical aspects of character.

Argument 8. Schematic legibility supplies an explanatory standard

Claim. If the constitutive identity is true, the organization can become readable through stable component semantics, compositional inference, localized counterfactual faults, and independent validation.

Rationale. A theory should make a trained reader able to reconstruct why a state is classified, which role fails under a perturbation, and where the inference becomes unlicensed.

Validation route. Test inter-rater reproducibility, fault prediction, role-preserving transport, estimator convergence, and compression from detailed causal data to a discriminating role profile.

Residual pressure. Structural intelligibility does not by itself bridge the metaphysical explanatory gap or prove the identity.

Argument 9. Constrained substrate independence avoids opposite extremes

Claim. The model permits multiple realization while rejecting both material chauvinism and consciousness from abstract similarity.

Rationale. A qualifying implementation must realize directionality, effective coupling, temporal grain, dynamic ranges, role structure, and substrate-appropriate viability.

Validation route. Establish construct, causal-role, and calibration preservation across substrates; use intervention and replacement tests rather than behavioral imitation alone.

Residual pressure. Nonbiological viability and cross-substrate phenomenal anchors remain underdeveloped.

The validation program at a glance

Phase	Scientific aim	Success criterion	What success does not yet establish
A. Synthetic construct separation	Test whether V, N_1, N_2, N_3 can vary and be measured separately	Recovery of known Boolean and continuous profiles under controls	Consciousness of the synthetic system
B. Near-miss atlas	Find natural and clinical diagnostic cases	Valid single-failure and grouped-failure coverage with complete audit records	Causal necessity without intervention
C. Selective perturbation and rescue	Change one realization while preserving retained roles	Licensed selectivity, predicted anchor change, and component-specific rescue	Metaphysical identity from one domain
D. Held-out transport	Test generality across subjects, tasks, modalities, and substrates	Stable prediction without favorable retuning	Universal substrate independence
E. Adversarial adjudication	Compare full and rival models under shared rules	Prospective outcome categories and independent implementation	Finality of the theory landscape

Quick objection-and-reply matrix

Objection	Best available reply	Residual pressure to acknowledge
The theory is just IIT + GNWT + RPT	The novelty is the exposed conjunction, distinct roles, viability, boundary order, conflict policy, and component audit	It must produce discriminating predictions not already forced by the source theories

Objection	Best available reply	Residual pressure to acknowledge
N_2 is access consciousness in disguise	Internal causal modulation and reduced-family tests exclude report and executive necessity	Implementation may still depend on access-like consumers
V is unfalsifiable	Freeze indicators and thresholds before the verdict and require concrete interpretability consequences	Repeated bespoke exceptions would weaken the theory
Thresholds are arbitrary	Make thresholds explicit, calibrated, cross-validated, and stress-tested	Brittle threshold dependence remains a real weakness
The model confuses correlates with constitution	Prioritize perturbation, route interruption, selective ablation, and rescue	Causal relevance still underdetermines metaphysical identity
It ignores qualia	It brackets presence first and adds a separate causal-geometry bridge for character	The bridge remains incomplete and symmetry-limited
It overattributes consciousness to AI	Mere behavior and graph isomorphism fail without causal realization, viability, boundary, and transport	Artificial-system viability and anchors remain contested
There is no oracle for C_0	Use independent convergent anchors and rival prediction, not self-generated labels	Conclusions remain anchor-relative
Consciousness may be graded	Keep continuous measures, thresholded scientific outputs, and a possible probabilistic extension	A sharp biconditional may ultimately need revision
Embodiment and affect are constitutive	Let boundary, V , dynamics, and character include organismic roles, and add a conjunct if evidence requires it	The current theory may understate life and value
The hard problem remains	The theory is a mid-level constitutive and empirical program, not a derivation of why experience exists	Philosophical dissatisfaction is not dissolved
Indeterminacy protects the model	Preregister all license and conflict rules and report unresolved rates as test properties	Excessive unresolved output would reduce usefulness

Part IV. Reproducible Course Materials

Appendix A. Corpus-to-Course Coverage Audit

This matrix lets the instructor verify that the course incorporates the conceptual contributions of the full paper corpus rather than relying only on the central formula.

Source	Concept or commitment	Primary treatment
Core paper	Minimal phenomenal presence distinguished from report, attention, and C_1	Sessions 1-2
Core paper	Necessary, sufficient, conjunction, and biconditional logic	Session 2; Problem Set 1
Core paper	Causal graph; structural, functional, and effective connectivity	Session 2
Core paper	Temporal grain and whole-brain / subcortical scope	Sessions 2, 5, 6
Core paper	N_1 integration or synergy and candidate surrogates	Session 4
Core paper	N_2 availability separated from report and fixed anatomy	Sessions 2 and 5
Core paper	N_3 bounded recurrent dwell and metastability	Session 6
Core paper	Independent viability and rescue-clause guardrail	Session 4
Core paper	Surrogate-conflict rule, preregistration, and robustness	Sessions 2 and 7
Core paper	Sufficiency, necessity, and exclusion-event falsifiers	Sessions 2 and 13
Core paper	Theory landscape and constrained substrate independence	Sessions 1, 12, 13
Core paper	Deliberate limitation to presence and initial quality-space proposal	Sessions 1 and 9
Boundary paper	Boundary relativity and boundary reversal	Session 3
Boundary paper	Physical, causal, and subject boundaries	Session 3
Boundary paper	Prohibition on retrospective score-maximizing boundaries	Session 3
Boundary paper	Endogenous capacity, inbound dependence, and autonomy ratio	Session 3
Boundary paper	Candidate reliability versus boundary stability	Session 3; Appendix D worksheet
Boundary paper	Role completeness and confidence-bound decisions	Sessions 3 and 7
Boundary paper	Inclusion-minimality and cross-scale hierarchy	Session 3
Boundary paper	Pairwise separability for multiple candidates	Sessions 3 and 11

Source	Concept or commitment	Primary treatment
Boundary paper	Constituent, interface, enabling support, and replacement test	Sessions 3 and 12
Boundary paper	Five boundary statuses	Sessions 3 and 7
Boundary paper	Split brain, brain-body, coupled agents, organoid-computer, distributed AI	Sessions 11-12
Availability paper	Activation / availability, availability / report, broadcast / diffusion	Session 5
Availability paper	Globality / extent, diversity / node count, posterior-subcortical / executive	Session 5
Availability paper	Independent recipient classes from causal repertoires	Session 5
Availability paper	Preparation burden and context robustness	Session 5
Availability paper	Content fidelity, class-native effects, and latency	Session 5
Availability paper	Selectivity and route-interruption evidence	Session 5
Availability paper	Coverage, class count, and diversity evenness	Session 5; Problem Set 2
Availability paper	Report- and executive-excluded recomputation	Session 5
Availability paper	Positive, negative, indeterminate, and unlicensed N_2 outputs	Sessions 5 and 7
Indeterminacy paper	Assessment tuple and validity envelope	Session 7; Appendix D worksheet
Indeterminacy paper	Four-outcome precedence rule	Session 7; Problem Set 2
Indeterminacy paper	Positive and Negative License Principles	Sessions 7 and 11
Indeterminacy paper	Five-part cause profile	Session 7; Appendix D worksheet
Indeterminacy paper	Surrogate-discordance rule and model-output sets	Session 7
Indeterminacy paper	Boundary and transport indeterminacy	Sessions 3, 7, 12
Indeterminacy paper	Coverage, indeterminate rate, unlicensed rate, selective risk	Session 7
Indeterminacy paper	Scientific status separated from practical action	Sessions 7 and 12
Ablation paper	Logical, incremental empirical, and causal necessity	Session 8
Ablation paper	Diagnostic-set theorem and approximate diagnostic region	Session 8; Problem Set 2
Ablation paper	Diagnostic coverage and underidentification	Session 8
Ablation paper	Hypothesis-not-definition and two evidence streams	Session 8
Ablation paper	Development, calibration, and test splits; no dual use	Sessions 8 and 14
Ablation paper	Held-out anchor risk and Δ_k	Session 8
Ablation paper	Criterion, mechanism, and measurement ablation	Session 8

Source	Concept or commitment	Primary treatment
Ablation paper	Selectivity license and rescue	Sessions 4, 6, 8
Ablation paper	Indispensable, redundant, coupled, indeterminate, unlicensed	Session 8
Ablation paper	Grouped ablation and weak versus strong minimality	Session 8
Ablation paper	Synthetic, near-miss, perturbation, transport, adversarial phases	Sessions 8, 13, 14
Character paper	Presence gate versus character map and two-layer error matrix	Session 9
Character paper	Perturbation-derived causal-trajectory geometry G_c	Session 9
Character paper	Independently anchored quality space	Session 9
Character paper	Held-out bridge F_Φ , nuisance controls, and stability	Session 9
Character paper	Geometry-changing intervention and deformation prediction	Session 9; Capstone options
Character paper	Character as an automorphism equivalence class	Session 9
Character paper	Modality, location, intensity, valence, temporal path	Sessions 6 and 9
Character paper	Color, audition, olfaction, pain, temporality, cross-modal scenes	Session 9; optional presentations
Character paper	Supported, refuted, indeterminate, unlicensed bridge outcomes	Session 9
Schematic paper	Black-box versus structurally legible necessity	Session 10
Schematic paper	Bearer, operation, unity, system-level presence, temporal presence role vector	Session 10
Schematic paper	Decomposable semantics, compositional readout, fault localization, independence	Session 10
Schematic paper	Correlational marker, black-box law, constitutive schematic	Session 10
Schematic paper	Observation of occurrence versus inference of character versus undergoing	Session 10
Schematic paper	Limits: representation, confirmation, theory-ladenness, scope, presence-character split	Session 10
Schematic paper	Role-preserving transport, fault prediction, convergence, compression, teachability	Sessions 10 and 13

Appendix B. Assessment Bank and Answer Guides

Problem Set 1. Concepts and arguments

1. Define minimal phenomenal presence and distinguish it from access consciousness and phenomenal character.
2. Translate $V \wedge (N_1 \wedge N_2 \wedge N_3) \Leftrightarrow C_0$ into two directional claims. Give a threat to each.
3. Explain why a boundary reversal is scientifically important even when neither candidate boundary is yet known to be correct.
4. Give one example each of activation without availability and availability without current activation.
5. Explain why the absence of a positive marker is not automatically a negative consciousness result.
6. State the diagnostic profile for auditing N_3 and explain why ordinary waking-versus-anesthesia data may not identify N_3 necessity.
7. Distinguish criterion, mechanism, and measurement ablation.
8. Explain why a high-performing neural manifold is not automatically a phenomenal quality space.
9. Apply D , C , F , and E to a one-number consciousness classifier.
10. State the strongest version of the “hybrid without novelty” objection and the best reply available in the paper corpus.
11. Explain how scientific indeterminacy can justify precaution without becoming a positive attribution.
12. Name one empirical result that should cause removal, combination, or addition of a conjunct.

Answer guide for Problem Set 1

1. C_0 is the minimal what-it-is-like fact. Access concerns usability for reasoning or control; character concerns what the admitted experience is like.
2. Sufficiency: the viable bundle guarantees C_0 ; threatened by licensed $V \wedge N^* \wedge \neg C_0$. Necessity: C_0 requires the viable bundle; threatened by licensed $C_0 \wedge \neg (V \wedge N^*)$.
3. It shows that the consciousness result depends on what was counted as the system and therefore requires independent individuation and sensitivity reporting.
4. Activation without availability: common-drive or arousal recruitment. Availability without activation: an idle recipient shown by nearby, low-burden intervention to be selectively reachable.
5. A negative requires applicability, sensitivity, adequate quality, satisfied auxiliary demands, a rule-out region, and exclusion of alternative causes of non-detection.
6. $V=1, N_1=1, N_2=1, N_3=0$ with uncertainty bounds clearing all gates. If all components move together across waking and anesthesia, full and ablated models agree on every case.
7. Criterion removes a rule term; mechanism changes a proposed physical realization; measurement removes an estimator or data route.
8. It may encode stimulus or task relations, occur unconsciously, or fit only correlationally. The bridge requires the presence gate, independent phenomenal anchors, held-out prediction, and intervention-sensitive deformation.
9. Stable semantics may be absent, the intermediate role profile is absent, faults are not localized, and validation may be circular. A label alone is not a schematic.
10. Objection: all components are borrowed. Reply: the exposed conjunction and its boundary, viability, report-independent, conflict, ablation, and revision rules make a distinct empirical architecture. Residual pressure: it must deliver unique predictions.

11. Scientific status describes evidential warrant; action also depends on stakes, reversibility, and error costs. Protective action does not alter D .
12. Examples: stable equivalence of a leave-one-out rival supports removal; inseparable components support combination; repeated conscious diagnostic cases with one condition absent support deletion or replacement; a new reliably necessary factor supports addition.

Problem Set 2. Formal classification practicum

Case A: Four-outcome decision

A test is applicable and well executed. Two credible surrogates disagree beyond the preregistered tolerance. The validated aggregate score would otherwise exceed τ_+ . What is the outcome and why?

Case B: Boundary status

Two disjoint candidates both clear autonomy and role thresholds. Both are stable across specifications. Interrupting either candidate's specific coupling to the other leaves the other's role profile intact; pairwise separability is .89 with $\kappa = .80$. What boundary status is supported? What does it not establish?

Case C: N_2 aggregation

Five independently calibrated recipient classes have frozen weights (.25, .20, .20, .20, .15). Classes 1, 2, and 4 pass every effect, latency, selectivity, route, and context gate. Class 3 is effect-indeterminate. Class 5 fails route dependence. Let $\alpha = .60$ and $m = 3$. The three qualifying effects are sufficiently even to pass η . Report and executive classes are not among the qualifying three. Is N_2 positive, negative, or indeterminate?

Case D: Diagnostic coverage

A study contains 500 cases. For N_1 , only four cases have the upper N_1 bound below threshold while lower bounds for V, N_2, N_3 clear their gates. The full model predicts their anchor patterns slightly better than the ablated model, but the confidence interval spans the minimum meaningful effect. What component status is warranted?

Case E: Mechanism selectivity

An intervention lowers N_3 by the preregistered amount but also lowers N_2 beyond its equivalence margin and changes the selected boundary. Independent anchors decline. What may be inferred?

Case F: Character bridge

Presence is positive. The bridge predicts baseline similarity well but fails held-out geometry-changing interventions, while the intervention and phenomenal-anchor validity checks pass. What is the bridge outcome and what remains intact?

Answer guide for Problem Set 2

- A. Indeterminate. Credible conflict has precedence over scalar aggregation.
- B. Multiple candidate bearers. The result identifies multiple stable candidates for a later consciousness test; it does not prove that either is conscious.
- C. Positive on the stated gates. Qualifying weight is .65, class count is three, diversity passes, and reduced-family requirements are satisfied. The unresolved class remains reported.
- D. Indeterminate and coverage-limited. Very low diagnostic coverage and an interval spanning the effect margin do not establish indispensability.
- E. The intended selective N_3 mechanism ablation is unlicensed; the result may reveal coupling or boundary reorganization. Anchor decline cannot be assigned specifically to N_3 .

F. Refuted for the specified bridge and domain. Positive presence remains intact; another character mapping may still succeed.

Problem Set 3. Essay and oral-exam prompts

1. Is CØ / N* a genuinely new theory or a disciplined synthesis? Defend a position using at least four commitments not entailed by simply listing IIT, GNWT, and RPT.
2. Does report-independent availability escape the phenomenal/access distinction, or does it relocate access inside the minimal condition?
3. Evaluate inclusion-minimality as a subject criterion. Compare it with IIT-style exclusion and an embodied alternative.
4. Can independent anchors test a consciousness theory without becoming a second consciousness theory?
5. Does the phenomenal-character extension answer the inverted-spectrum objection or formalize its remaining force?
6. Does schematic legibility explain consciousness, or only make a functional identity easier to understand?
7. Design one licensed negative test in a nonreporting population and defend every negative-license condition.
8. Should the sharp biconditional be replaced with a probabilistic model? State what empirical pattern would justify the revision.

Analytic-writing rubric

Criterion	Excellent	Competent	Needs revision
Target clarity	Exact consciousness target and scope are maintained throughout	Target is stated but occasionally drifts	"Consciousness" is used without qualification
Theory accuracy	Roles, order, and outcome rules are reconstructed correctly	Minor omissions do not change the argument	Major components or licenses are conflated
Rival fairness	Strong, version-specific rival and objection	Rival is recognizable but simplified	Strawman or target mismatch
Evidence discipline	Boundary, validity, anchors, confounds, and transport are addressed	Most licenses are addressed	Evidence is equated with the target
Falsifiability	Concrete result and revision rule are stated	General counterevidence is named	No outcome could change the view
Argument quality	Premises, inference, residual pressure, and conclusion are explicit	Coherent argument with limited residual analysis	Assertion or summary without inference

Appendix C. Capstone Preregistration and Rubric

Capstone title and team

Study title:

Team members and declared prior expectations:

Planned registration, data, and code repository:

1. Exact target

- Target form of consciousness:
- Candidate content or state:
- Population or substrate:
- Claim deliberately excluded from the study:

2. Full model and rivals

State the full $C\emptyset / N^*$ prediction and at least one serious rival. The rival may be a leave-one-out conjunction, grouped ablation, probabilistic version, source theory, nuisance model, or nonconjunctive theory.

Model	Prospective prediction	Expected anchor pattern	Revision if disfavored
Full model			
Rival 1			
Rival 2, if used			

3. Diagnostic region

- Exact cases on which the models disagree:
- Expected diagnostic coverage and how it will be estimated:
- Continuous diagnostic margin and uncertainty rule:
- Off-manifold or sampling concerns:

4. Candidate-system specification

- Observational universe Ω and known omissions:
- Proposed candidates and discovery-only generator:
- Grain g , interval Δ , and operating regime R :
- Autonomy estimators, thresholds, and uncertainty bounds:
- Inclusion-minimality, separability, and stability rules:
- Constituent, interface, and enabling-support tests:

5. Component operationalization

Component	Estimator family	Threshold and uncertainty	Main confound	Negative control
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V

Component	Estimator family	Threshold and uncertainty	Main confound	Negative control
N_1				
N_2				
N_3				

For N_2 , additionally state recipient-class calibration, content and matched control, admissible contexts, preparation burden, effect and latency thresholds, selectivity and route tests, class weights, diversity rules, and report/executive exclusions.

6. Intervention and selectivity

- Target mechanism or criterion:
- Matched sham, energy, timing, and nuisance controls:
- Minimum intended component change:
- Equivalence margins for every retained component:
- Boundary and viability preservation checks:
- Rescue or recovery condition:
- Conditions that make the intervention unlicensed:

7. Independent anchors

- Anchor vector and target relation of each element:
- Validation genealogy and known auxiliary demands:
- Prohibited dual uses:
- Leave-family-out and shared-channel sensitivity analysis:
- Preregistered translation from each model to expected anchor patterns:

8. Evidence separation

Partition	Permitted use	Prohibited use
Development	Choose candidate estimators and interventions	Final model adjudication
Calibration	Freeze thresholds, bridge rules, and uncertainty	Outcome-driven tuning
Final test	Score frozen predictions once	Any redesign except labeled exploration

9. Outcome rules

Define each outcome before critical data access.

Outcome	Exact rule	Scientific interpretation	Next study
Positive / support			
Negative / challenge			
Indeterminate			

Outcome	Exact rule	Scientific interpretation	Next study
Unlicensed			
Coupled or redundant, if applicable			

10. Robustness and transport

- Reasonable specification neighborhood:
- Boundary, grain, estimator, threshold, and anchor variants:
- Multiplicity control:
- Transport target and three bridge requirements:
- Minimum result stability:
- Maximum acceptable unresolved-output rate:

11. Practical action note

If the study bears on clinical or welfare decisions, state possible actions, error costs, reversibility, and precaution independently of the scientific classification.

12. Stopping and revision commitments

- Sample-size and stopping rule:
- Exclusion and missingness policy:
- Model change after each possible result:
- Result that would cause abandonment of the favored interpretation:
- Exploratory analyses permitted after confirmatory completion:

Capstone rubric

Dimension	Points	Full-credit standard
Target and scope	10	Exact explanandum, system, content, regime, and excluded claims
Rival quality	10	Strong rivals with genuinely different prospective predictions
Diagnostic identification	12	Correct disagreement set, coverage plan, and uncertainty margin
Boundary discipline	10	Independent proposal, stability, minimality, and separability addressed
Construct validity	12	Component measures and confounds are explicitly validated
Anti-circularity	12	Evidence splits, anchor independence, and no dual use are enforceable
Intervention selectivity	10	Retained components, identity, viability, and rescue are protected
Outcome discipline	10	Positive, negative, indeterminate, unlicensed, and special statuses are prospective
Robustness and transport	8	Reasonable alternatives and transfer limits are declared
Revision and oral defense	6	Every result has a substantive consequence and objections are answered directly

Appendix D. Reusable Instructor Worksheets

Worksheet 1. Complete case-classification record

Field	Entry
Target form of consciousness	
Candidate system, boundary status, and omitted variables	
Grain, regime, and interval	
Independent viability result	
N_1 result and surrogate agreement	
N_2 result and reduced-family checks	
N_3 result and temporal interpretation	
Assessment applicability and data quality	
Final outcome	
Cause profile $\{e, s, m, b, t\}$	
Most informative next study	
Separate practical-action note	

Worksheet 2. Theory-comparison grid

Question	$C\emptyset$ / N^*	Rival theory
Exact explanandum		
Candidate bearer		
Necessary condition(s)		
Claimed sufficient condition		
Evidence treated as constitutive		
Report and access role		
Phenomenal-character account		
Substrate commitment		
Unique diagnostic case		
Direct falsifier		

Worksheet 3. Boundary audit

Step	Required record
Observational universe	Included variables, known omissions, interfaces
Specification	Grain, horizon, regime, intervention family

Step	Required record
Candidate proposal	Discovery-only method and candidate family
Autonomy	J_{self}, J_{in} , ratio, uncertainty, reliability
Role completeness	V, N_1, N_2, N_3 for every frozen candidate
Minimality	Qualifying proper subsets
Separability	Reciprocal interruption and preservation
Replacement	Constituent, interface, or support
Stability	Cross-specification mapping and result
Boundary status	Unique, multiple, nested, none, or indeterminate

Worksheet 4. Component-necessity audit

Question	Entry
Component under audit	
Full and ablated model	
Exact diagnostic profile	
Diagnostic coverage	
Component estimator and threshold	
Independent anchor vector	
Development / calibration / test split	
Dual-use check	
Mechanism selectivity, if claimed	
Full versus rival held-out risk	
Specification stability	
Status: indispensable, redundant, coupled, indeterminate, unlicensed	

Worksheet 5. Phenomenal-character bridge

Field	Entry
Positive presence-gate evidence	
Content family and causal trajectories	
Geometry type, distance, transitions, occupancy	
Independent phenomenal anchors	
Train / validation / test split	
Null and nuisance models	
Held-out metric and transition performance	
Geometry-changing intervention	

Field	Entry
Deformation prediction and observed change	
Surviving automorphisms	
Transport license	
Bridge outcome	

Worksheet 6. Schematic-legibility check

Condition	Evidence
Decomposable semantics	Roles are stable and do not repeat the outcome label
Compositional readout	Intermediate organization profile can be reconstructed
Counterfactual fault localization	Different changes yield different predicted failures
Evidential independence	Readout is not used to validate its own identity claim
Teachability	Independent readers agree on held-out cases
Transport	The role profile survives materially different realizations when appropriate

Appendix E. Glossary and Notation

Term or symbol	Teaching definition	Guardrail
C_0 or $C\emptyset$	Minimal phenomenal presence	Not report, intelligence, wakefulness, or full character
C_1	Enriched reflective or metacognitive consciousness	Not built into the minimal gate
N^*	$N_1 \wedge N_2 \wedge N_3$	A defined bundle; its identity with consciousness is a hypothesis
V	Independently assessed capacity to realize the relevant dynamics	Not biological life alone and not a post hoc rescue clause
N_1	Integration or synergy	Not activity, correlation, or complexity in general
N_2	Counterfactual system-wide availability	Not report, diffusion, anatomical extent, or node count
N_3	Bounded recurrent stability	Not mere lateness, persistence, or memory
S	Boundary-qualified candidate system	Must be frozen before theory-specific scoring
c	Target content or content-bearing state	Results can be content-relative
g	Spatial, functional, and temporal grain	Cross-grain comparisons need a mapping
R	Operating regime	State changes can alter validity and boundaries
Δ	Evaluation horizon or interval	Presence is windowed rather than instantaneous
Effective connectivity	Directed causal influence among modeled variables	Not anatomy or statistical dependence alone
Synergy	Joint information or causal contribution irreducible to parts	Estimator families require construct validation
Recipient class	Functionally distinct group defined by causal response repertoire	Not an anatomical label or parcel count
Preparation burden Γ	Cost of placing a recipient in a nearby test context	Prevents installing new access machinery
Content selectivity	Effect beyond energy-matched and content-destroying controls	Broad response alone is diffusion-compatible
Route dependence	Loss of a content-specific effect under matched route interruption	Redundant routes require adequate family coverage
C_A	Weighted coverage across qualifying recipient classes	Weights cannot simply reward node count
K_A	Number of qualifying recipient classes	Prevents one class from passing alone
H_A	Evenness of effect across qualifying classes	Prevents token threshold crossings around one dominant class
N_2^{-R}	Availability recalculated without report classes	Report remains optional evidence
N_2^{-E}	Availability recalculated without executive classes	Executive uptake cannot be the sole pass route

Term or symbol	Teaching definition	Guardrail
Validity envelope V	Domain in which an evidence-target interpretation is established	A formula does not carry its meaning automatically
Positive	Evidence licenses attribution of the exact target	Not certainty or complete explanation
Negative	Valid, sensitive, adequate evidence licenses rule-out	Not failure to cross a positive threshold
Indeterminate	Valid evidence does not discriminate enough	Not half conscious and not invalid
Unlicensed	Interpretation conditions failed	Meta-output, not evidence of absence
Cause profile	Evidential, surrogate, model, boundary, and transport limitations	Not a consciousness score
Surrogate conflict	Credible measures exceed preregistered disagreement tolerance	Must be checked before aggregation
Boundary reversal	Result changes when the candidate boundary changes	Shows that individuation is part of the claim
J_{self}	Candidate's endogenous causal capacity	Blocks inert independence
J_{in}	Inbound external imposition on organization	Does not penalize outbound agency
A_{θ}	Ratio of endogenous capacity to endogenous plus inbound dependence	Eligibility for candidacy, not consciousness amount
Inclusion-minimality	No qualifying proper subsystem at the same specification	Not fewest nodes or highest score
Separability	Each candidate preserves organization under reciprocal channel interruption	Low separability does not prove one larger subject
Constituent	Specific dynamics are required after generic support replacement	More than remote causal necessity
Interface	Bounded input or output channel compatible with autonomous role completeness	Causal contact alone does not make it internal
Enabling support	Generic replacement restores the role profile	Time-horizon dependent
Diagnostic set D_k	Cases where X_k fails and every retained component passes	Only region where full and leave-one-out conjunctions differ
Diagnostic coverage Γ_k	Target-distribution mass in the diagnostic region	A rare corner limits generality
Criterion ablation	Remove a term from the decision rule	Tests incremental prediction, not mechanism
Mechanism ablation	Intervene on a proposed realization	Requires selectivity and preserved identity
Measurement ablation	Remove an estimator or data route	Tests robustness, not physical necessity
Anchor vector Y	Independent convergent evidence used for model comparison	Not an infallible consciousness oracle
Causally coupled	Components cannot be varied separately within the tested intervention family	Motivates composite modeling or better intervention
Weak minimality	No single component can be removed	Can miss grouped redundancy
Strong minimality	Full model beats every preregistered proper submodel	Still relative to anchors, domain, and rival class

Term or symbol	Teaching definition	Guardrail
G_c	Content's neighborhood in a causal-trajectory geometry	Not a snapshot or arbitrary latent embedding
Q	Independently anchored empirical quality-space structure	Need not be Euclidean or globally commensurable
F_ϕ	Frozen bridge from causal-geometry features to quality structure	High in-sample correlation is insufficient
Q_ϕ	Symmetry-qualified phenomenal-character map	Applied only after a positive presence gate
$Aut(Q)$	Transformations preserving every measured quality relation	Records residual inversion and label underdetermination
Π_c	Temporal path features of an admitted character	Distinct from N_3 's presence gate
Schematic legibility	Publicly teachable compositional and counterfactual readout	Intelligibility is not proof
Role vector ρ_{C_0}	Bearer, operation, unity, system-level presence, temporal presence	The intermediate explanatory profile
Transport	Preservation of construct, causal role, and calibration across domains	Surface similarity is insufficient

Appendix F. Primary Corpus and Selected Contextual Readings

Primary CØ / N* corpus

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- Stilwell, P. (2026). *Reading consciousness from the schematic: Structural legibility in the N model** [Manuscript].

Project site: <https://philstilwell.github.io/C0/>

Selected contextual readings

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Closing instructor note

The deepest pedagogical virtue of this corpus is not the compactness of $V \wedge N^* \Leftrightarrow C_0$. It is the insistence that every inference earn its target, bearer, evidence, scope, and revision rule. If students finish the course able to preserve those distinctions while designing a study that could surprise both proponents and critics, the manual has done its job.